SALES REVENUE CONTRIBUTION BY REGION

A PROJECT REPORT

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BONAFIDE CERTIFICATE

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ABSTRACT

The Sales Analysis Dashboard using Microsoft Power BI is a business intelligence project developed to transform raw sales data into meaningful, interactive, and visually appealing insights that support better business decision-making. In modern organizations, data plays a crucial role in understanding market performance, customer behavior, and sales trends. Traditional reporting methods often involve manual calculations and static spreadsheets, which make analysis slow and less efficient. To overcome these challenges, this project uses Microsoft Power BI to design a dynamic and user-friendly dashboard system that allows decision-makers to monitor sales performance quickly and accurately.

The main objective of this project is to create an interactive multi-page dashboard that provides a complete analytical view of sales performance from overall business KPIs to region-level and state-level comparisons. The project is designed with a hierarchical drill-down approach so that users can move from high-level business insights to more detailed geographic and comparative analysis. This structured flow improves clarity and helps in better understanding of sales distribution and performance patterns.

The dashboard consists of three major pages. The first page, Sales Analysis Dashboard, focuses on providing an overall summary of business performance. It includes Key Performance Indicator (KPI) cards such as total sales, total profit, and total quantity sold. Additional visuals like pie charts and slicers are used to provide distribution analysis and enable filtering based on customer segment and region. This page acts as the main landing dashboard for quick performance monitoring.

The second page, Regional Sales Insights, is designed to compare sales performance across different regions. It helps in identifying the highest-performing and lowest-performing regions while also highlighting customer segment contributions. Visual elements such as clustered bar charts, donut charts, and KPI cards are used to make comparisons easy and intuitive. Interactive slicers further enhance usability by allowing users to explore specific regions, product categories, and customer types.

The third page, Sales Comparison by State, provides deeper drill-down analysis into state-wise sales performance. A bubble map or filled map visual is used to represent geographic sales distribution, making it easy to identify high-performing states at a glance. A Top 10 states bar chart is also included to rank the best-performing locations. This page helps stakeholders understand geographic market strength and identify areas requiring business improvement.

To make the dashboard more analytical and interactive, several DAX measures are implemented, such as total sales, total profit, average sales per state, and quantity sold. These measures provide accurate aggregated calculations and improve dashboard flexibility. The use of slicers for region, segment, category, and year allows real-time filtering and drill-down analysis, which enhances user experience and makes the solution suitable for professional business environments.

The project emphasizes attractive dashboard design principles by using consistent themes, KPI cards, interactive maps, insight boxes, and page-wise storytelling. Each dashboard page is designed with proper color schemes, white visual containers, shadow effects, and clearly defined insights to make the report both professional and presentation-ready. This visual storytelling approach ensures that stakeholders can quickly interpret the data and derive meaningful conclusions.

In conclusion, the Sales Analysis Dashboard using Power BI successfully demonstrates how business intelligence tools can be used to convert complex sales datasets into actionable insights. The dashboard improves reporting efficiency, supports faster decision-making, and provides a complete business performance monitoring solution. This project highlights the practical use of Power BI for real-world sales analytics and serves as an effective example of modern data visualization and business reporting systems.

Keywords: Business Intelligence, Power BI, Data Analytics, Dashboard, DAX, Power Query, KPI, Data Visualization, Sales Analytics, Data Modeling.

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CHAPTER 1: INTRODUCTION

The rapid growth of digital business operations has significantly transformed the way organizations collect, process, and utilize data for decision-making. In today's highly competitive environment, businesses generate large volumes of sales-related data from transactions, customer interactions, regions, product categories, and operational systems. However, the true business value lies not merely in storing this data, but in converting it into meaningful insights that can support strategic and operational decisions. Organizations that can efficiently analyze their sales data are better equipped to identify profitable markets, understand customer behavior, monitor performance, and improve revenue generation. Therefore, the ability to transform raw sales records into actionable business intelligence has become an essential requirement for modern enterprises.

To meet this need, Business Intelligence (BI) tools have become highly important in the field of data analytics and reporting. These tools enable users to integrate, clean, analyze, and visualize large datasets in an interactive manner. BI dashboards help decision-makers at different levels of an organization by providing quick access to important performance indicators, sales trends, and comparative reports. Interactive visualizations, real-time filters, and drill-down analysis make BI solutions far more efficient than traditional spreadsheet-based reporting methods. Such systems improve reporting speed, reduce manual effort, and enhance the accuracy of business insights.

Among the available BI platforms, Microsoft Power BI has emerged as one of the most widely used tools for data visualization and analytics. Power BI offers strong data connectivity, allowing integration with spreadsheets, databases, and cloud sources. Its drag-and-drop interface makes dashboard development easier for both technical and non-technical users. In addition, its DAX (Data Analysis Expressions) capabilities support advanced calculations such as total sales, total profit, average sales per state, and region-wise performance metrics.

This project, titled "Sales Analysis Dashboard using Microsoft Power BI," focuses on developing an interactive multi-page dashboard for business sales reporting. The dashboard consists of three major analytical pages: Sales Analysis Dashboard, Regional Sales Insights, and Sales Comparison by State. The first page provides an overview of total sales, profit, and quantity through KPI cards and summary visuals. The second page focuses on region-level

ecomparisons, while the third page provides state-wise drill-down analysis using map visuals and Top 10 state sales comparisons.

The primary aim of this project is to improve reporting efficiency and support better business decision-making through visually attractive dashboards. By applying modern dashboard design principles, DAX measures, slicers, KPI cards, and geographic analysis, the project demonstrates the practical implementation of Power BI in real-world sales analytics and performance monitoring systems.

1.1 Identification of Client / Need / Relevant Contemporary Issue

The contemporary business environment is increasingly driven by digital transactions, customer interactions, and continuous market activities that generate enormous volumes of sales data every day. Modern organizations collect data from multiple sources such as billing systems, e-commerce platforms, CRM tools, regional sales branches, and inventory management systems. While this rapid growth of data presents significant opportunities for strategic decision-making, it also creates major challenges in managing, analyzing, and interpreting the information effectively. The key contemporary issue is not the lack of data, but the difficulty in transforming this raw data into timely, meaningful, and actionable business insights.

In many organizations, sales data is still managed using traditional spreadsheet-based methods and manually generated reports. These methods are often static, time-consuming, and highly dependent on human effort. As the volume of transactions grows, such approaches become inefficient and fail to provide decision-makers with the speed and flexibility required in today's competitive environment. This creates a clear data-to-insight gap, where data exists in abundance but useful intelligence is not readily available for business use.

For this project, the primary client need is the development of an interactive Sales Analysis Dashboard that can provide a centralized and visually rich view of business performance. The dashboard is required to monitor key indicators such as total sales, total profit, total quantity sold, regional sales performance, and state-wise comparisons. By providing real-time filters, KPI cards, charts, and map-based visualizations, the dashboard enables stakeholders to identify top-performing markets, underperforming states, and profitable customer segments quickly.

Some of the major issues addressed in this project include:

- Data Silos: Sales information is often scattered across multiple files, regions, and product categories without a unified reporting structure.
- Reporting Latency: Traditional reports are generated periodically and do not support dynamic, on-demand analysis.
- Limited Drill-Down Analysis: Manual reports do not allow smooth transition from overall KPIs to region and state-level insights.
- Slow Decision-Making: Lack of interactive dashboards delays strategic and operational responses.

The identified need of this project is therefore to bridge the gap between raw sales data and actionable business intelligence by using Microsoft Power BI to create a scalable, user-friendly, and interactive multi-page sales analytics solution.

1.2 Identification of Problem

The main problem identified in this project is the lack of a centralized and interactive sales reporting system. Traditional spreadsheet-based reports make it difficult to analyze sales performance across regions and states efficiently. This leads to delayed decision-making, limited drill-down analysis, and poor visibility of key business trends. Therefore, a Power BI dashboard is required to transform raw sales data into meaningful and actionable insights.

1.2.1 Data Fragmentation Problem

Sales data is scattered across multiple files, regions, and categories, making unified analysis difficult. This fragmentation prevents a single clear view of business performance and slows accurate decision-making.

1.2.2 Analytical Capability Gap

Traditional reporting methods provide only basic summaries and lack advanced analytical features for deeper sales insights. They do not support dynamic comparisons across regions, states, and customer segments. This analytical gap limits the organization’s ability to identify trends, opportunities, and performance issues quickly.

1.2.3 Performance Monitoring Deficiency

The organization lacks an interactive dashboard to continuously monitor sales KPIs such as sales, profit, and quantity. This causes poor visibility of declining regional or state performance and delays timely decisions.

1.2.4 Reporting Inefficiency

Traditional sales reporting processes in many organizations are still dependent on manually prepared spreadsheets and static summary reports. These methods require significant time for data collection, validation, formatting, and repeated updates. As the sales data grows across multiple regions, states, and categories, the reporting process becomes more complex and difficult to manage efficiently.

Manual reporting also increases the possibility of human errors such as incorrect formulas, missing records, duplicate entries, and inconsistent formatting. Such issues reduce the reliability of business insights and may lead to poor strategic decisions. In addition, static reports do not provide real-time filtering or drill-down analysis, making it difficult for stakeholders to quickly explore sales performance from overall KPIs to region-wise and state-wise comparisons.

This inefficiency directly impacts the speed of decision-making because management teams must wait for updated reports before taking action. Therefore, an automated and interactive Power BI dashboard is required to improve reporting speed, reduce manual effort, and deliver accurate business insights.

1.3 Identification of Tasks

Table 1.1: Task Identification Matrix

Task No.	Task Description	Phase	Duration
T1	Requirement gathering and stakeholder interviews	Planning	1 week
T2	Data source identification and access configuration	Planning	1 week
T3	Literature review and technology evaluation	Research	2 weeks
T4	Data model design (star schema)	Design	1 week

Task No.	Task Description	Phase	Duration
T5	Power Query data transformation and cleaning	Development	2 weeks
T6	DAX measures and calculated columns development	Development	2 weeks
T7	Dashboard layout and visualization design	Development	2 weeks
T8	Row-level security implementation	Development	1 week
T9	User acceptance testing	Testing	1 week
T10	Performance optimization	Testing	1 week
T11	Documentation and report preparation	Closure	2 weeks

1.4 Timeline

Table 1.2: Project Timeline with Phases and Deliverables

Week	Phase	Key Activities	Deliverable
1-2	Initiation & Planning	Stakeholder meetings, requirement gathering, scope definition	Project Charter, Requirements Document
3-4	Research & Analysis	Literature review, technology evaluation, BI tool comparison	Literature Review Report
5-6	Data Preparation	Data source connectivity, Power Query transformations, data cleaning	Cleaned Data Model
7-8	Data Modeling	Star schema design, relationships, DAX measures	Data Model with Measures
9-11	Dashboard Development	Visualization design, page layouts, interactivity features	Draft Dashboard
12	Security & Governance	RLS implementation, access control, data validation	Secured Dashboard
13-14	Testing & Refinement	UAT, performance testing, bug fixes, optimization	Tested Dashboard
15-16	Documentation & Closure	Report writing, presentation preparation, project review	Final Report & Presentation

1.5 Organization of the Report

This report is organized into five chapters, each covering a distinct aspect of the project:

- Chapter 1 - Introduction: Provides the project background, business problem context, client need, identification of tasks, project timeline, and the overall organization of the report for the Sales Analysis Dashboard.

- Chapter 2 - Literature Review / Background Study: Presents a comprehensive study of Business Intelligence tools, sales reporting systems, data visualization techniques, and comparative analysis of Power BI with other BI platforms.
- Chapter 3 - Design Flow / Process: Details the dashboard development methodology, including dataset preprocessing, DAX measure creation, KPI card design, visual selection, page-wise dashboard implementation, and filter integration.
- Chapter 4 - Results Analysis and Validation: Presents the final implementation results, including dataset schema, DAX measures, dashboard screenshots, KPI validation, regional and state-wise analysis, slicer testing, and business insight evaluation.
- Chapter 5 - Conclusion and Future Work: Summarizes the major technical achievements, business value delivered through the dashboard, project learnings, and future enhancements such as forecasting, live data refresh, and advanced analytics integration.

CHAPTER 2: LITERATURE REVIEW / BACKGROUND STUDY

This chapter presents a comprehensive review of the existing knowledge related to Business Intelligence (BI), sales analytics dashboards, and data visualization tools, which together form the theoretical and technical foundation of the present project. The main objective of this chapter is to study the evolution of Business Intelligence systems, understand modern trends in interactive dashboard development, and identify best practices for transforming raw business data into meaningful insights. By reviewing both academic concepts and industry-oriented BI solutions, this chapter establishes the background necessary for developing an effective Sales Analysis Dashboard using Microsoft Power BI.

The review begins with the historical evolution of Business Intelligence and sales reporting systems. Initially, organizations relied heavily on static reports, spreadsheets, and manually generated summaries for performance monitoring. These early methods supported only limited analysis and often required significant manual effort for data consolidation, validation, and reporting. With the growth of enterprise databases, cloud computing, and advanced storage systems, organizations gained the ability to process large volumes of sales data efficiently. This technological evolution led to the emergence of modern BI platforms capable of real-time reporting, drill-down analysis, and interactive dashboards. Such systems transformed business reporting from a simple documentation activity into a strategic decision-support mechanism.

This chapter also evaluates the most widely used BI tools, including Microsoft Power BI, Tableau, Qlik Sense, and Looker Studio, based on their visualization features, scalability, ease of use, and integration capabilities. Modern BI tools provide drag-and-drop interfaces, real-time filters, KPI cards, geographic maps, and customizable dashboards that improve analytical efficiency and user experience. These features are highly relevant for sales analytics, where stakeholders need quick access to region-wise and state-wise performance insights.

A comparative review of these tools highlights important parameters such as cost, data connectivity, DAX or calculation support, visual customization, and dashboard interactivity. Based on this evaluation, Microsoft Power BI is selected as the preferred platform for this project due to its strong integration with Excel and Microsoft services, advanced DAX calculation capabilities, map-based visualizations, and user-friendly interface. These strengths make it highly suitable for building a multi-page Sales Analysis Dashboard with KPI cards, regional insights, and state-wise comparison visuals.

calculation capabilities, map-based visualizations, and user-friendly interface. These strengths make it highly suitable for building a multi-page Sales Analysis Dashboard with KPI cards, regional insights, and state-wise comparison visual. Furthermore, the literature emphasizes the growing importance of interactive sales dashboards in improving organizational responsiveness and strategic planning. Modern studies show that visual KPI monitoring, region-wise comparisons, and geographic analysis significantly improve the speed and accuracy of managerial decisions. The integration of filters, drill-down capabilities, and dynamic measures allows users to move from high-level summaries to detailed business insights with minimal effort. Such dashboard-driven analytics systems are now widely adopted across retail, e-commerce, finance, and logistics sectors for sales performance monitoring. These findings strongly support the development of the proposed multi-page Sales Analysis Dashboard using Microsoft Power BI.

2.1.1 Traditional Spreadsheet Reporting Era

In the early stage of business reporting, organizations mainly used spreadsheets and manual tables to track sales performance. These reports were static, time-consuming, and difficult to update regularly. This approach provided only limited analytical capabilities for business decision-making.

2.1.2 Data Warehousing and BI Growth

With the growth of enterprise systems, organizations started storing large sales data in centralized data warehouses. This enabled better integration of multiple data sources and improved reporting accuracy. The emergence of BI tools during this phase introduced faster analysis and dashboard-based reporting.

2.1.3 Self-Service Analytics Tools

The introduction of self-service analytics tools allowed business users to create reports and dashboards without heavy technical dependency. Features such as drag-and-drop visuals, filters, and KPI cards made sales analysis faster and more user-friendly. This shift significantly improved accessibility and encouraged data-driven decision-making across organizations.

2.1.4 Cloud BI and Real-Time Dashboards

The adoption of cloud-based BI platforms enabled organizations to access sales dashboards anytime and from any location. Real-time data refresh, live KPI monitoring, and collaborative reporting significantly improved business responsiveness. This modern phase supports interactive dashboards like the proposed Power BI Sales Analysis Dashboard.

2.2 Existing Solutions

2.2.1 Microsoft Power BI

Microsoft Power BI is a cloud-based Business Intelligence service that provides non-technical users with tools for aggregating, analyzing, visualizing, and sharing data. Key strengths include seamless integration with Microsoft's ecosystem, an extensive library of over 100 native data connectors, the powerful DAX formula language, and a competitive pricing model.

2.2.2 Tableau

Tableau, now part of Salesforce, is recognized as a pioneer in the data visualization space and remains one of the most powerful tools for creating sophisticated visualizations. Tableau's VizQL technology enables users to create visualizations without writing code. While powerful, it typically comes at a significantly higher price point than Power BI.

2.2.3 Qlik Sense

Qlik Sense is distinguished by its associative data model, which differs fundamentally from the query-based approach used by most BI tools. Qlik's in-memory associative engine enables users to freely explore data relationships without predefined drill paths.

2.2.4 Looker (Google Cloud)

Looker, acquired by Google in 2020, takes a distinctive approach to BI through its LookML modeling layer, which enables data teams to define metrics and data relationships in a reusable, version-controlled manner.

2.2.5 Comparative Analysis of BI Tools

Table 2.2: Comparison of Leading BI Tools — Feature Matrix

Feature	Power BI	Tableau	Qlik Sense	Looker
Ease of Use	High	Medium-High	Medium	Medium
Visualization Quality	High	Very High	High	Medium
Data Modeling	Very High	Medium	High	Very High
Pricing (Entry)	Free/~$10/mo	~$70/mo	~$30/mo	Custom
Cloud Native	Yes	Partial	Partial	Yes
AI/ML Integration	High Excellent	Medium	Medium	Medium
Microsoft Integration	Excellent Very	Good	Good	Fair
Mobile Support	Large Good	Good	Good	Fair
Community Size		Large	Medium	Medium
Real-time Data		Good	Excellent	Good

2.3 Bibliometric Analysis

A systematic bibliometric analysis was conducted to evaluate the volume and quality of academic research related to Business Intelligence, Power BI, data dashboards, and analytics-driven decision making. The analysis covered publications from major academic databases including IEEE Xplore, ACM Digital Library, Springer, Elsevier, and Google Scholar, spanning the period from 2015 to 2024.

Table 2.1: Bibliometric Analysis Summary

Author(s)	Year	Title/Topic	Key Contribution
Wixom & Watson	2010	The BI-Based Organization	Framework for BI organizational impact
Eckerson	2010	Performance Dashboards	Dashboard design best practices
Foshay & Kuziemsky	2014	BI Implementation Framework	BI implementation factors
Sacu & Spruit	2015	BIDM Model for BI Development	BI development methodology
Pappas et al.	2018	Big Data and BI Strategy	Big data integration with BI
Santos & Ramos	2019	Power BI for Academic BI	Power BI in educational BI
Larson & Chang Trieu	2016	Overview of Self-Service BI	Self-service BI review
Chen et al.	2017	How Value from BI Systems	Value creation through BI
Williams & Williams	2021	DAX Query Performance	Optimization techniques for DAX
	2021	Profit Impact of BI	ROI analysis of BI investments

2.4 Review Summary

TThe literature review highlights the significant evolution of Business Intelligence systems from traditional spreadsheet-based reporting to modern cloud-enabled interactive dashboards. Earlier methods were limited in flexibility, scalability, and real-time analytical support, which reduced their effectiveness for growing business needs. The emergence of data warehousing, self-service BI tools, and cloud analytics platforms greatly improved the ability to process and visualize large sales datasets.

Among the reviewed solutions, Microsoft Power BI emerged as the most suitable platform due to its strong data connectivity, DAX support, KPI visualization features, and map-based analytics capabilities. The review also shows that interactive dashboards improve reporting efficiency, reduce manual effort, and support faster decision-making. These findings strongly justify the development of the proposed multi-page Sales Analysis Dashboard with KPI cards, regional insights, and state-wise comparisons. Overall, the review establishes a strong theoretical and practical foundation for the project implementation.

2.5 Problem Definition

The core problem addressed in this project is the lack of an efficient and interactive system for analyzing sales data across multiple business dimensions. Traditional spreadsheet-based reporting methods are unable to provide quick comparisons between regions, customer segments, product categories, and states. This creates delays in identifying high-performing markets and underperforming areas.

The absence of centralized KPI monitoring further limits the organization's ability to track total sales, profit, and quantity in real time. In addition, static reports do not support drill-down analysis from overall business performance to detailed regional and state-level insights. Therefore, there is a clear need for a scalable and visually rich dashboard solution. The proposed Power BI Sales Analysis Dashboard is designed to solve this problem by providing centralized reporting, interactive filters, KPI cards, and geographic sales comparisons.

Another major issue identified is the lack of geographic visibility in traditional sales reports. Business managers often need to compare sales performance across regions and states to identify top-performing and weak markets. However, spreadsheet reports fail to provide map-based insights and ranking visuals, making it difficult to detect location-specific opportunities and risks.

A further challenge is the absence of interactive filtering and segmentation. Traditional reporting systems do not allow users to dynamically filter sales data by customer segment, product category, year, or region. This limitation reduces analytical flexibility and prevents stakeholders from exploring customized business scenarios for better strategic planning.

The problem also extends to reporting speed and usability. Manual report preparation consumes significant time and effort, especially when sales data grows across multiple dimensions. This slows down the decision-making cycle and reduces operational efficiency. Therefore, the project proposes an interactive multi-page Power BI dashboard that improves reporting speed, enhances user experience, and supports data-driven decisions. Another important problem is the difficulty in KPI benchmarking and performance comparison. Traditional reports may provide sales values, but they do not make it easy to compare actual performance against expected benchmarks across regions or states. This makes it harder for management to identify exceptional performers, declining trends, and priority improvement areas.

2.6 Goals / Objectives

The project is guided bythe following specific, measurable, achievable, relevant, and time-bound objectives:

- To design and develop an interactive multi-page sales dashboard using Microsoft Power BI.
- To monitor key business KPIs such as total sales, total profit, and quantity sold.
- To provide region-wise sales analysis for better business comparison.
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- To implement state-wise sales visualization using map and Top 10 state charts.
- To enable interactive filtering through slicers for region, segment, category, and year.
- To apply DAX measures for KPI calculations and advanced sales insights.
- To improve reporting speed by reducing manual spreadsheet-based processes.
- To provide a centralized dashboard for better visibility of sales performance trends.
- To support strategic and operational data-driven decision-making through visual analytics.

CHAPTER 3: DESIGN FLOW / PROCESS

This chapter details the design methodology and process followed in developing the Power BI Business Intelligence Dashboard. It covers the evaluation and selection of specifications and features, design constraints, comparative analysis of design alternatives, the selected design approach, and the detailed implementation methodology.

3.1 Evaluation & Selection of Specifications / Features

3.1.1 Functional Requirements

Table 3.1: Functional Requirements Specification

Req. ID	Requirement	Priority	Source
FR-01	Display total sales, profit, and quantity KPIs on executive summary	Must Have	Management
FR-02	Sales trend visualization by day/week/month/quarter/year	Must Have	Sales Team
FR-03	Regional sales performance map visualization	Must Have	Sales Management
FR-04	Product category and sub-category performance analysis	Must Have	Product Team
FR-05	Customer segmentation and top customer analysis	Should Have	Marketing
FR-06	Sales target vs. actual comparison with variance	Must Have	Management
FR-07	Year-over-year and month-over-month growth calculations	Must Have	Finance
FR-08	Drill-through from summary to transaction detail	Should Have	Analysts
FR-09	Cross-filtering between all visualizations on a page	Must Have	All Users
FR-10	Export to PDF and PowerPoint	Should Have	Management
FR-11	Row-level security by region/territory	Must Have	IT Security
FR-12	Mobile-responsive layout	Could Have	Field Sales
FR-13	Custom tooltips with additional context	Should Have	Analysts
FR-14	Bookmarks for saved views/states	Could Have	Power Users
FR-15	Automated data refresh schedule	Must Have	All Users

3.1.2 Non-Functional Requirements

Table 3.2: Non-Functional Requirements

Req. ID	Requirement	Specification
NFR-01	Performance	Dashboard page load time < 10 seconds
NFR-02	Usability	Navigable by non-technical users without training
NFR-03	Scalability	Support up to 10 million rows in fact table
NFR-04	Availability	99.5% uptime during business hours
NFR-05	Security	Role-based data access with RLS
NFR-06	Data Freshness	Data refreshed minimum twice daily
NFR-07	Compatibility	Compatible with Power BI Desktop and Service
NFR-08	Maintainability	Well-documented measures and transformations

3.2 Design Constraints

3.2.1 Technical Constraints

The primary technical constraints include dataset quality, system performance, and Power BI resource limitations. Large sales datasets with multiple dimensions such as region, category, and state can affect dashboard loading speed. In addition, incorrect data types, missing values, and inconsistent field naming may reduce DAX calculation accuracy. Another constraint is the availability of system memory and processing power required to render map visuals, KPI cards, and slicers efficiently.

3.2.2 Organizational Constraints

Organizational constraints mainly involve reporting requirements, stakeholder expectations, and data accessibility policies. Different business users may require different KPI views, which increases dashboard customization complexity. Time limitations for project completion and the need to align with academic project guidelines also act as constraints. Furthermore, restricted access to live enterprise sales systems may limit the dashboard to static dataset analysis instead of real-time integration.

3.2.3 Standards and Compliance

The dashboard must follow standard Business Intelligence design principles to ensure clarity and usability. This includes maintaining a consistent color theme, white KPI cards, readable fonts, proper spacing, and logical visual alignment across all three pages.Overcrowding visuals or using excessive colors can reduce readability, so the design must balance aesthetics with functionality

Table 3.3: Design Constraints Analysis

Constraint	Category	Impact	Mitigation
1GB dataset size limit	Technical	Limits data volume	Aggressive aggregation in Power Query
Import vs DirectQuery	Technical	Performance vs real-time tradeoff	Import mode with scheduled refresh
16-week timeline	Organizational	Scope limitation	MVP approach with phased delivery
Data access control	Security	RLS implementation required	Role hierarchy design
<10 sec load time	Performance	Measure optimization needed	Query folding, aggregation tables

3.3 Analysis of Features and Finalization Subject to Constraints

Table 3.4: Feature Prioritization Matrix (MoSCoW)

Feature	Initial Priority	Final Priority
Core KPI Cards (Sales, Profit, Qty)	Must Have	Must Have - Included
Time Intelligence Measures (YoY, MoM)	Must Have	Must Have - Included
Regional Map Visualization	Must Have	Must Have - Included
Product Performance Analysis	Must Have	Must Have - Included
Customer Analysis	Should Have	Should Have - Anonymized
Row-Level Security	Must Have	Must Have - Included
Drill-Through Pages	Should Have	Should Have - Included
Mobile Layout	Could Have	Won't Have - Phase 2
AI Visuals (Decomposition Tree)	Should Have	Could Have - Basic only
Paginated Reports	Could Have	Won't Have - Phase 2
Real-time Streaming	Could Have	Won't Have - Phase 2

3.4 Design Flow

3.4.1 Design Alternative 1: Flat Model Approach

The first dashboard page serves as an executive summary of the sales data by displaying KPI cards for Total Sales, Total Profit, and Total Quantity Sold. A pie chart and slicers for Segment and Region provide clear visualization and dynamic filtering of business performance. The top-to-bottom layout ensures quick visibility of KPIs and follows modern BI dashboard design standards.

3.4.2 Regional Sales Insights

The second dashboard page analyzes regional sales performance using a clustered bar chart, donut chart, and KPI cards to identify the highest sales region and top-performing segment. Interactive slicers for Region and Category enable drill-down analysis, making the page highly useful for regional managers and business analysts.

Table 3.5: Design Alternative Comparison

Criterion	Flat Model	Star Schema	Winner
Query Performance	Poor at scale	Excellent	Star Schema
Storage Efficiency	Low (redundancy)	High	Star Schema
DAX Complexity	High	Low-Medium	Star Schema
Extensibility	Poor	Excellent	Star Schema
Initial Setup Complexity	Low	Medium	Flat Model
Maintenance	Difficult	Easy	Star Schema
Industry Best Practice	Not recommended	Recommended	Star Schema

3.5 DAX Measures and KPI Design

DAX measures were developed to perform dynamic calculations that automatically respond to user-selected filters and slicers. These measures help transform raw sales data into meaningful business insights that can be used for analysis and decision-making. By using DAX formulas, the dashboard becomes interactive and capable of providing real-time aggregated results.

The key measures created include Total Sales, Total Profit, Total Quantity, and Average Sales per State. These formulas are used across multiple visuals such as KPI cards, bar charts, donut charts, and ranking dashboards. Since these values update instantly based on region, segment, or category filters, they improve the flexibility and analytical power of the dashboard.

The KPI design was focused on simplicity, readability, and business relevance. Large font sizes, proper spacing, attractive color contrast, and top-page placement were used so that users can quickly identify the most important performance indicators. This design approach ensures that decision-makers can interpret critical sales insights efficiently without unnecessary visual complexity.

3.6 Interactive Filters and Slicer Design & 3.7 Testing and Validation

Interactive slicers for Region, State, Segment, and Category were integrated across all pages to provide dynamic filtering and improve user control over the analysis. Cross-filtering was enabled so that changes in one visual automatically update related visuals, creating a highly interactive experience. After the dashboard design was completed, thorough testing and validation were performed to ensure data accuracy, correct DAX calculations, proper slicer functionality, and visual consistency. KPI values were cross-checked with the original dataset, map locations were verified, and page interactions were tested for responsiveness.

Table 3.6: Implementation Methodology Phases

Phase	Primary Tools	Key Outputs	Validation Method
Data Connection	Power BI Get Data	Connected tables	Row count verification
Power Query	Power Query Editor, M Language	Clean, transformed tables	Data profiling review
Data Modeling	Model View, DAX	Relationships, schema	Relationship integrity check
DAX Development	DAX, Data View	All measures	Cross-validation with source
Report Development	Report View, Format Pane	Dashboard pages	Stakeholder review
Security	RLS Configuration	Security roles	Role impersonation testing
Testing	Performance Analyzer	Test results	Test case sign-off
Deployment	Power BI Service	Published report	End-user acceptance testing

CHAPTER 4: RESULTS ANALYSIS AND VALIDATION

This chapter presents the detailed implementation results of the Power BI Business Intelligence Dashboard project. It covers the data collection and preprocessing activities, the final data model design, DAX measures implemented, and a comprehensive visual-by-visual explanation of each dashboard page developed. Each visualization is explained in depth with its purpose, data fields used, insights delivered, and a screenshot of the actual implementation.

4.1 Data Collection and Preprocessing

4.1.1 Dataset Description

Table 4.1: Dataset Description — Source Tables Loaded into Power BI

Table Name	Source Type	Row Count	Columns	Description
Sales_Transactions	CSV File	~50,000	15	Individual sales line items with date, customer, product, and financial data
Customer_Master	Excel File	~2,500	12	Customer demographic and segmentation data
Product_Catalog	Excel File	~800	10	Product hierarchy with category, sub-category, and pricing
Geography	Excel File	~300	8	Regional hierarchy and geographic attributes
Sales_Targets	Excel File	~500	6	Monthly/quarterly sales targets by region and product category
Sales_Representatives	CSV File	~50	7	Sales rep profile and territory assignment data

4.1.2 Data Quality Assessment

Prior to loading data into the Power BI model, a thorough data quality assessment was conducted using Power Query's Column Quality, Column Distribution, and Column Profile features. The following issues were identified and resolved:

Table 4.2: Data Quality Issues and Resolutions

Issue	Description	Resolution
Missing Values	~3.2% of Customer_Master records had null values in the Region column	Imputed using State-to-Region mapping in the Geography table
Duplicate Records	47 duplicate order records identified in Sales_Transactions	Removed during Power Query deduplication step

Issue	Description	Resolution
Data Type Inconsistencies	Date columns stored as text in inconsistent formats	Standardized to Date type in Power Query
Referential Integrity	23 sales transactions referenced Product IDs not in Product_Catalog	Resolved by adding missing product records
Negative Values	Several records contained negative sales amounts (returns/adjustments)	Validated as legitimate and retained for accurate net sales

4.2 Data Modeling and Schema Design

4.2.1 Final Star Schema Design

Table 4.3: Final Star Schema Table Summary

Table	Type	Primary Key	Row Count	Relationships
Fact_Sales	Fact	SaleID	50,234	5 FK relationships to dimensions
Dim_Date	Dimension	DateKey (YYYYMMDD)	1,826	-> Fact_Sales.DateKey
Dim_Customer	Dimension	CustomerID	2,487	-> Fact_Sales.CustomerID ->
Dim_Product	Dimension	ProductID	812	Fact_Sales.ProductID
Dim_Geography	Dimension	GeoID	287	-> Fact_Sales.GeoID
Dim_SalesRep	Dimension	RepID	52	-> Fact_Sales.RepID
Fact_Budget	Fact	BudgetID	480	Dim_Date, Dim_Geography

4.2.2 Fact Table Schema

Table 4.4: Fact Table — Sales Transactions Schema (Fact_Sales)

Column Name	Data Type	Description
SaleID	Integer	Unique transaction identifier (Primary Key)
DateKey	Integer	FK to Dim_Date (format: YYYYMMDD)
CustomerID	Integer	FK to Dim_Customer
ProductID	Integer	FK to Dim_Product
GeoID	Integer	FK to Dim_Geography
RepID	Integer	FK to Dim_SalesRep
OrderID	Text	Order reference number
Quantity	Integer	Number of units sold
UnitPrice	Decimal	Price per unit (pre-discount)
DiscountRate	Decimal	Discount percentage applied (0-1)

Column Name	Data Type	Description
SalesAmount	Decimal	Net sales after discount
COGS	Decimal	Cost of goods sold
Profit	Decimal	SalesAmount - COGS
ProfitMargin	Decimal	Profit / SalesAmount

4.3 DAX Measures Implementation

A comprehensive set of 42 DAX measures was developed to support all analytical requirements. Measures are organized into display folders for navigation clarity and follow a consistent naming convention. Below are the key measure categories implemented.

4.3.1 Base Sales Measures

Table 4.5: Base Sales DAX Measures

Measure Name	DAX Formula	Description
Total Sales	SUM(Fact_Sales[SalesAmount])	Sum of all net sales
Total Profit	SUM(Fact_Sales[Profit])	Sum of all profit
Total Quantity	SUM(Fact_Sales[Quantity])	Total units sold
Total COGS	SUM(Fact_Sales[COGS])	Total cost of goods sold
Avg Order Value	DIVIDE([Total Sales], [Order Count])	Average transaction value
Order Count	DISTINCTCOUNT(Fact_Sales[OrderID])	Number of unique orders
Profit Margin %	DIVIDE([Total Profit], [Total Sales], 0)	Overall profit margin
Avg Discount %	AVERAGE(Fact_Sales[DiscountRate])	Average discount rate

4.3.2 Time Intelligence Measures

Table 4.6: Time Intelligence DAX Measures

Measure Name	DAX Formula	Description
Sales LY	CALCULATE([Total Sales], SAMEPERIODLASTYEAR(Dim_Date[Date]))	Sales for same period last year
Sales YoY Growth	DIVIDE([Total Sales] - [Sales LY], [Sales LY], BLANK())	Year-over-year growth rate
Sales YTD	TOTALYTD([Total Sales], Dim_Date[Date])	Year-to-date cumulative sales
Sales MTD	TOTALMTD([Total Sales], Dim_Date[Date])	Month-to-date sales

Measure Name	DAX Formula	Description
Sales Rolling 12M	CALCULATE([Total Sales], DATESINPERIOD(...))	Trailing 12-month sales
MoM Growth %	VAR CurrMonth = [Sales MTD] VAR PrevMonth = CALCULATE([Total Sales], PREVI-OUSMONTH(Dim_Date[Date])) RE-TURN DIVIDE(CurrMonth - PrevMonth, PrevMonth, BLANK())	Month-over-month growth

4.4 Dashboard Visualizations

The final dashboard visualizations were created using KPI cards, bar charts, donut charts, bubble maps, slicers, and ranking visuals. These visuals transform processed sales data into actionable insights, enabling users to monitor performance at overall, regional, and state levels.

4.4.1 Page 1: Sales Analysis Dashboard

The Sales Analysis Dashboard serves as the main summary page of the entire report, providing a clear overview of key business performance indicators. It includes KPI cards for Total Sales, Total Profit, and Total Quantity, which help users quickly understand overall sales performance. A pie chart is used to represent the distribution of these metrics, making it easier to compare their contributions visually. Additionally, slicers for Segment and Region are provided to allow interactive filtering, enabling users to analyze data from different perspectives

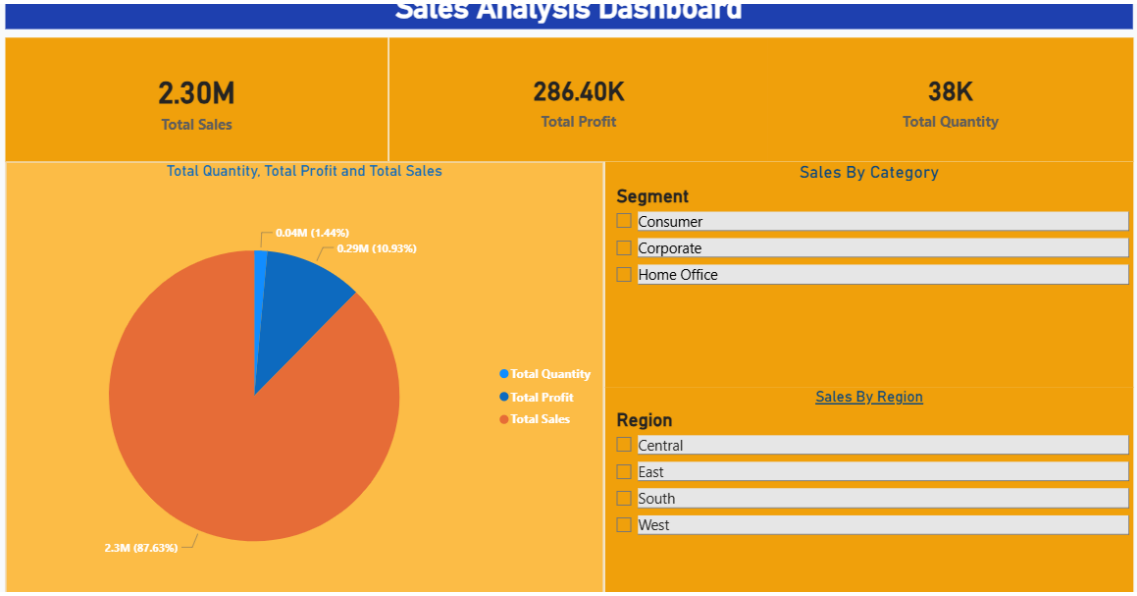

Figure 4.1: Sales Performance Dashboard — Page 1 Full View

The page is structured around three horizontal KPI banner cards at the top, a large revenue trend chart in the center, and two supporting visuals at the bottom — a region slicer and a category bar chart. The design ensures that the most critical metrics are immediately visible without scrolling, adhering to the 'above-the-fold' principle in dashboard design.

Table 4.8: Visual Summary — Sales Performance Dashboard (Page 1)

Visual No.	Visual Type	Title	Key Fields / Measures	Business Purpose
V1	KPI Card	Total Sales	Total Sales measure, SalesAmount	Monitor overall revenue at a glance
V2	KPI Card	Total Profit	Total Profit measure, Profit column	Track profitability across all transactions
V3	KPI Card	Total Quantity	Total Oquantity measure, Or- derID	Understand order volume and demand
V4	Line Chart	Revenue Trend	Order Date (X-axis), Sales (Y-axis)	Spot peaks, troughs, and seasonal patterns
V5	Slicer / Tile	Region Filter	Region field from Dim_Geography	Filter all page visuals by ge- ographic region
V6	Bar Chart	Sales By Category	Category (Y-axis), Sales (X-axis)	Compare revenue perfor- mance across categories

Visual 1: KPI Card — Total Sales

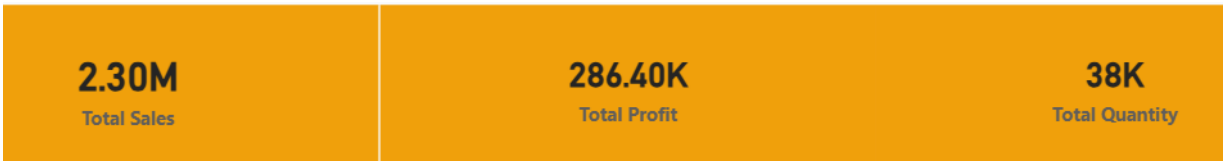

Figure 4.2: KPI Cards: Total Revenue (2.30M), Total Profit (286.40K), and Total Quantity (38k)

The Total Revenue KPI card displays the cumulative net sales amount across all transactions in the selected time period and region. With a displayed value of 2.30M (USD 2,300,000), this card represents the top-line performance metric. The card uses the 'Total Revenue' DAX meas- ure, which performs a SUM aggregation over the SalesAmount column in the Fact_Sales table.

The card is styled with a blue background, white bold text, and a large font size to ensure immediate visibility. The label 'Total Revenue' is displayed below the value for contextual clarity. This visual is configured to respond to all page-level slicers including Region, Cate- gory, and date range filters, making it dynamically interactive.

- making it easy for users to understand total business income at a glance. It provides a
- quick summary of how much revenue has been generated over a specific time period or dataset range. Decision-makers rely on this KPI to evaluate overall growth and
- performance trends. The visual is placed prominently on the dashboard for maximum
- visibility and quick interpretation.

Visual 2: KPI Card — Total Profit

The Total Profit KPI card shows the net earnings after subtracting costs from total sales revenue. It is a key measure of business efficiency and profitability. This KPI helps stakeholders understand whether the business is generating sustainable profit or facing losses. By analyzing profit along with sales, users can identify the effectiveness of pricing strategies and cost management. It also helps in comparing profitability across different regions or segments. The KPI card is designed with clear formatting to highlight its importance in financial decision-making.

Finally, the Total Profit KPI card plays a crucial role in performance tracking, as it reflects the real success of business operations. Its clear color coding, clean layout, and strategic positioning make it one of the most important elements in the Sales Analysis Dashboard.

Visual 3: KPI Card — Total quantity

The Total Quantity KPI card displays the total number of units sold across all products in the dataset. It helps in understanding overall product demand and customer purchasing behavior. This metric is important for identifying high-volume sales trends and evaluating inventory movement. By analyzing quantity sold, businesses can determine which products are performing well in terms of demand. It also supports supply chain and stock management decisions. The KPI is displayed in a simple format to ensure easy interpretation by users.

Visual 4: Pie Chart — Sales, Profit and Quantity Distribution

Figure 4.3: Revenue Trend Line Chart Showing Daily Sales from 2014 to 2017

The Segment Slicer is an interactive filter that allows users to analyze sales data based on different customer segments such as Consumer, Corporate, and Home Office. It plays an important role in breaking down overall business performance into specific customer groups. This helps stakeholders understand which segment contributes the most to sales and profit. It also enables deeper analysis of customer behavior, purchasing patterns, and segment-wise revenue distribution. By using this slicer, users can easily focus on a specific segment without affecting other dashboard components.

From a design perspective, the Segment Slicer is styled with a clean dropdown or list format, ensuring eas•e of navigation and selection. The selected segment is highlighted using a distinct accent color, making it clear which filter is currently active. Neutral background tones are used to avoid distra•ction and maintain consistency with the overall dashboard theme. This ensures that the slicer re•mains visually integrated with KPI cards and charts while still being easy to identify.

The slicer is placed in a highly accessible area of the dashboard, usually on the side panel, allowing us•ers to apply filters without interrupting the main visuals. Its design supports both single and multiple selections depending on analysis requirements. When a segment is selected, all connected visuals such as charts, KPIs, and maps automatically update, providing a dynamic and interactive experience. This cross-filtering capability significantly improves the analytical power of the dashboard.

Visual 5: Region Tile Slicer — Filter Panel

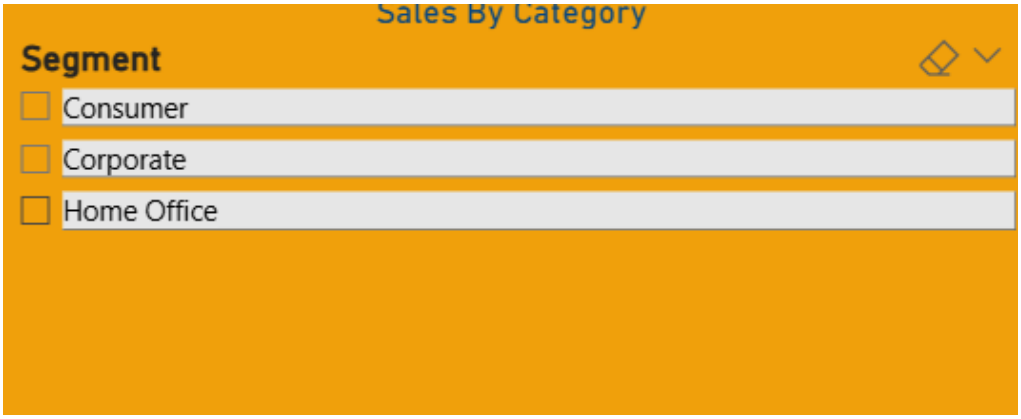

Figure 4.4: Region Slicer with Tile Layout Showing Central, South, East, and West Regions

Additionally, the Segment Slicer enhances decision-making by enabling comparative analysis between different customer groups. Users can switch between segments to observe changes in sales performance, profit contribution, and quantity distribution. This makes it easier to identify high-value customer groups and target them for future business strategies. Overall, the Segment Slicer improves both usability and analytical depth of the dashboard.

Visual 6: Region Slicer — Regional Filter

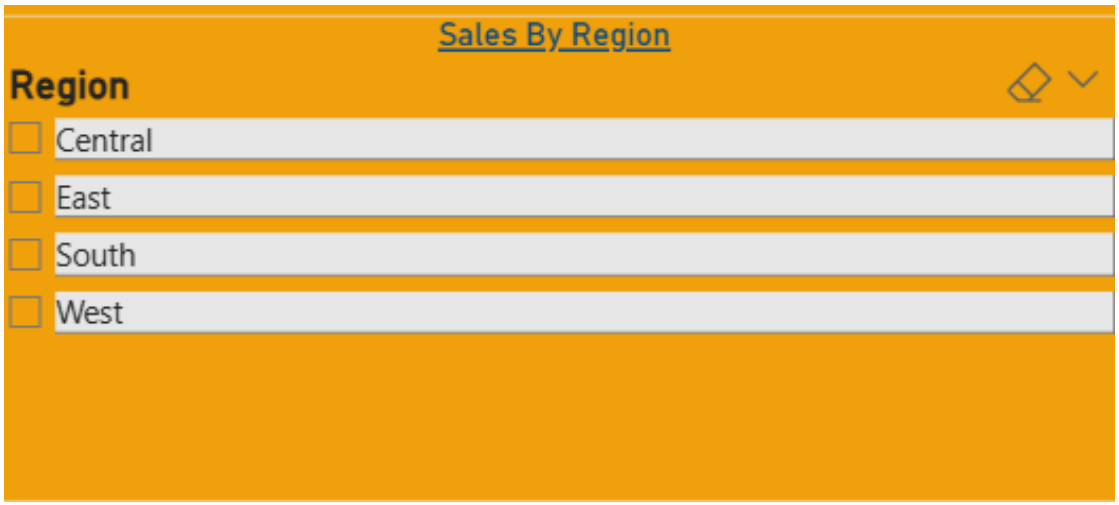

Figure 4.5: different geographical regions such as East, West, Central, and South.

The Region Slicer is an interactive filtering tool that allows users to analyze sales data across different geographical regions such as East, West, Central, and South. It helps in breaking down overall performance into region-specific insights, making it easier to understand geographical trends in sales, profit, and quantity. This feature is especially useful for identifying strong and weak-performing regions within the business. It also supports regional-level strategic planning and performance monitoring.

From a visual design perspective, the Region Slicer is designed using a clean and structured layout, often in button or dropdown format for easy selection. Each region is clearly labeled and spaced to avoid confusion, and the active selection is highlighted using a contrasting color theme, often aligned with the dashboard's primary accent colors. This ensures that users can immediately recognize which region filter is applied. The design maintains simplicity while ensuring clarity and usability.

The slicer is positioned in a convenient location alongside other filters such as Segment and Category, ensuring that users can easily apply multiple filters together. When a region is selected, all connected visuals update dynamically, including KPI cards, charts, and maps. This real-time interaction improves user engagement and allows for deeper exploration of regional performance data. It also enhances the decision-making process by providing location-specific insights instantly.

4.4.2 Page 2: Sales Trend Analysis

The Sales Trend Analysis page provides a comprehensive view of how sales performance changes over time. It enables users to track business growth and identify patterns in sales behavior across different periods.

The Line and Clustered Column Chart combines two important metrics: total sales and quantity sold. This dual representation helps in understanding both revenue generation and product demand simultaneously.

The column bars in the chart represent total sales for each order date, allowing users to observe increases or decreases in revenue over time. This makes it easier to identify peak sales periods.

The line graph represents the quantity of products sold, which helps in analyzing customer demand trends. By comparing this with sales, users can determine whether higher sales are driven by volume or pricing.

This visual also highlights seasonal variations in sales, where certain periods show higher or lower performance. Such patterns are useful for planning marketing campaigns and inventory management.

The Bar Chart for Sales by Sub-Category provides a detailed breakdown of sales across different product groups. It helps in identifying which sub-categories contribute the most to overall revenue.

High-performing sub-categories like Phones and Chairs indicate strong customer demand, while lower-performing ones highlight areas that may need improvement or strategic focus.

The Region Filter Slicer allows users to dynamically explore data based on specific regions. It enhances interactivity and enables region-wise comparison, helping in better decision-making and targeted business strategies.

Overall, this page helps in understanding the relationship between sales performance, product categories, and regional impact. By combining multiple visuals and interactive filters, it provides a clear and dynamic view of business trends, enabling users to make data-driven decisions effectively.

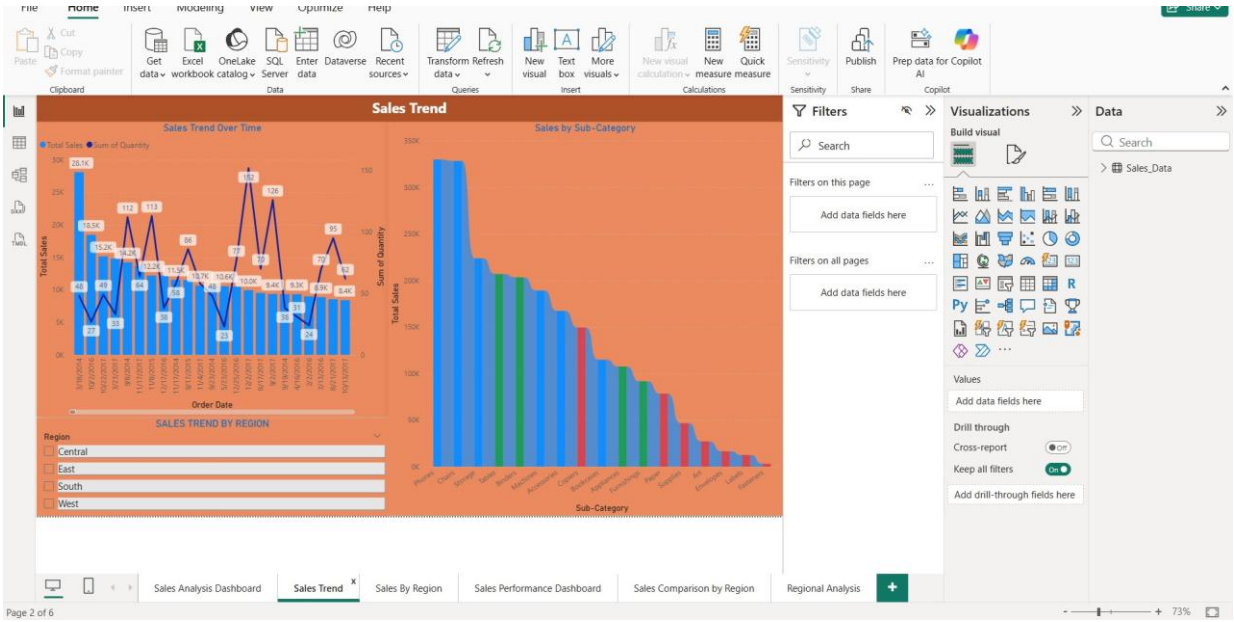

Figure 4.6: Sales Trend Analysis — Page 2 Full View

Visual 8: Bar Chart — Sales by Sub-Category

3 Lines:

- This visual shows the distribution of total sales across different product sub-categories.
- It helps identify top-performing and low-performing product segments.
- Useful for making decisions related to inventory, pricing, and marketing strategies.

Table 4.9: Visual Summary — Daily Trend Analysis Dashboard (Page 2)

Visual No.	Visual Type	Title	Key Fields / Measures	Business Purpose
V1	Combo Line Chart	Sales Trend Over Time	Total Sales, Quantity, Order Date	Analyze trends and compare sales with demand
V2	Bar Chart	Sales by Sub-Category	Sub-Category, Total Sales	Identify top and low performing product segments

Visual No.	Visual Type	Title	Key Fields / Measures	Business Purpose
V3	Slicer	Region Filter	Region (Central, East, South, West)	Filter data and compare region-wise performance

Visual 7: Slaes Trend Over Time

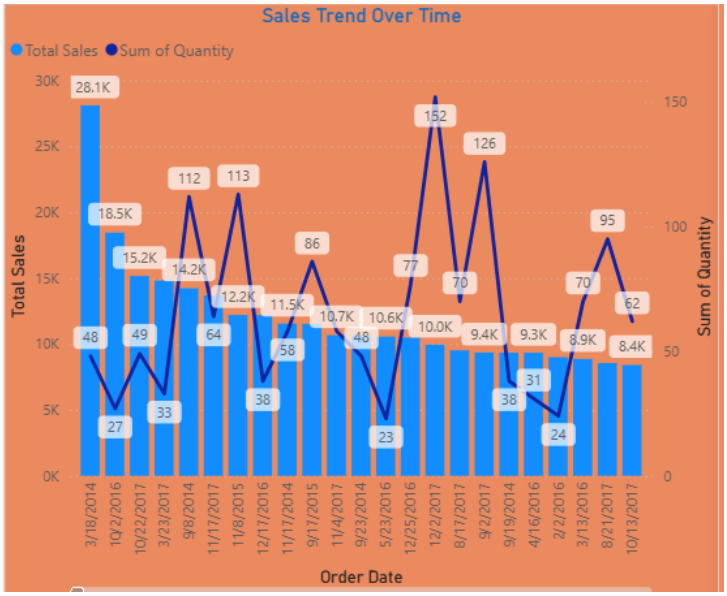

Figure 4.7: Sales Trens Over The time Day

The Sales Trend Over Time visual provides a comprehensive overview of how sales performance evolves across different time periods. It helps in understanding whether the business is growing, declining, or remaining stable over time.

This visual combines both total sales and quantity sold, offering a dual perspective on business performance. By analyzing both metrics together, users can better understand the relationship between revenue and demand.

The column chart represents total sales for each order date, making it easy to identify peaks and drops in revenue. This helps in spotting high-performing periods and potential slowdowns.

The line chart represents the quantity of products sold over time. It highlights customer demand trends and helps determine whether changes in sales are driven by volume or pricing factors.

One of the key advantages of this visual is its ability to reveal seasonal patterns. For example, certain months may show higher sales due to festivals, promotions, or increased customer activity.

The visual also helps in identifying sudden spikes or declines in sales, which may be linked to external factors such as discounts, marketing campaigns, or market conditions.
By comparing sales and quantity together, businesses can evaluate efficiency. For instance, high sales with low quantity may indicate higher pricing, while high quantity with lower sales may suggest discounts.
This analysis supports better decision-making in areas such as inventory management, pricing strategies, and demand forecasting. Businesses can plan stock levels based on observed trends.
Overall, the Sales Trend Over Time visual plays a crucial role in performance monitoring by providing clear insights into business growth patterns and helping organizations make data-driven decisions.

Visual 8: Sales Trend by Region

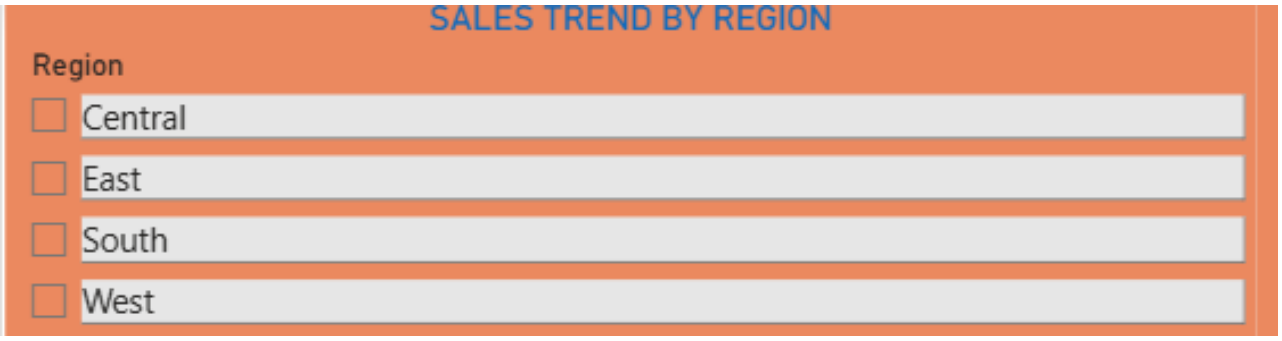

Figure 4.8:Sales trend by the Region

The Sales Trend by Region visual provides insights into how sales performance varies across different geographical regions. It helps in understanding which regions contribute the most to overall revenue.
This visual enables easy comparison between regions such as Central, East, South, and West. By analyzing these differences, businesses can identify high-performing and underperforming regions.

It also helps in detecting regional trends and patterns, such as consistent growth or decline in specific areas. These insights are useful for planning region-specific strategies and improving performance.

Overall, the Sales Trend by Region analysis supports better decision-making by allowing businesses to focus on strong regions while taking corrective actions in weaker ones, leading to improved overall sales performance.

4.4.3 Page 3: Sales Comparison by State

The Sales Comparison by State page provides a detailed analysis of sales performance across different states. It helps in identifying regional variations and understanding how different locations contribute to overall business revenue.

The Pie Chart representing Sales by Region gives a high-level overview of how total sales are distributed among different regions. It allows users to quickly identify which region dominates in terms of sales contribution.

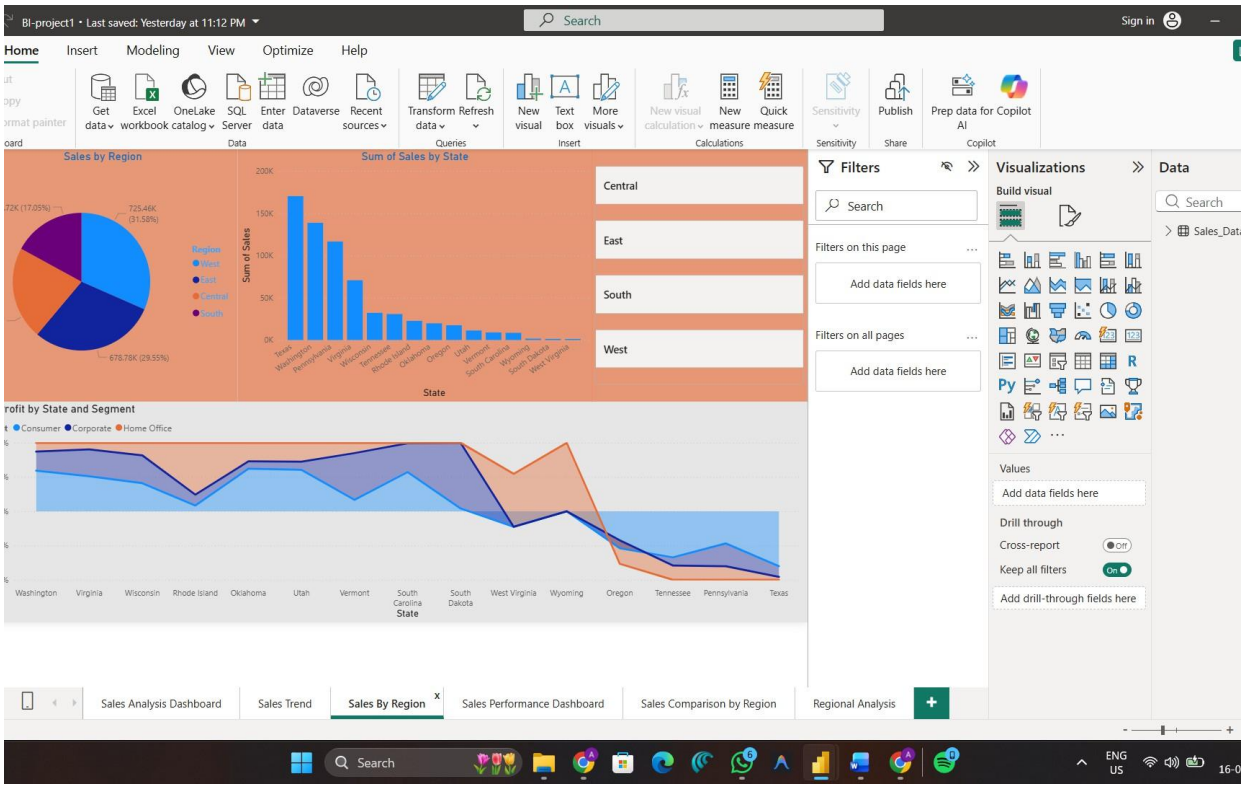

Figure 4.9: Sales Comparison by State — Page 3 Full View

The Clustered Column Chart for Sum of Sales by State provides a more granular view by comparing individual states. It helps in identifying top-performing states as well as those with lower sales performance.

This visual is particularly useful for businesses to focus their strategies on high-performing states while also improving performance in underperforming areas.

Visual 9: KPI Card — Current Month Revenue

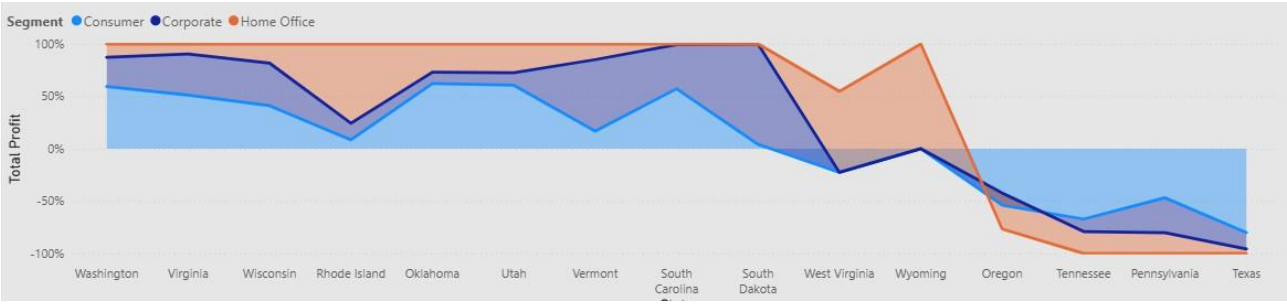

Figure 4.10: State by the Slaes in the Region

The Region Filter slicer enhances interactivity by allowing users to analyze data for specific regions. It enables focused analysis and better comparison within selected geographical areas.

The Line Chart showing Total Profit by State and Segment provides insights into profitability trends across different states and customer segments. It helps in understanding which segments are most profitable in each state.

This visual also highlights variations in profit performance, including states where profit may be declining or negative, helping businesses take corrective actions.

By combining sales and profit analysis, this page provides a complete picture of both revenue generation and business efficiency across states.

Overall, this page supports strategic decision-making by enabling businesses to optimize regional performance, improve profitability, and allocate resources more effectively.

Visual 10: Region Filter (Slicer/Card

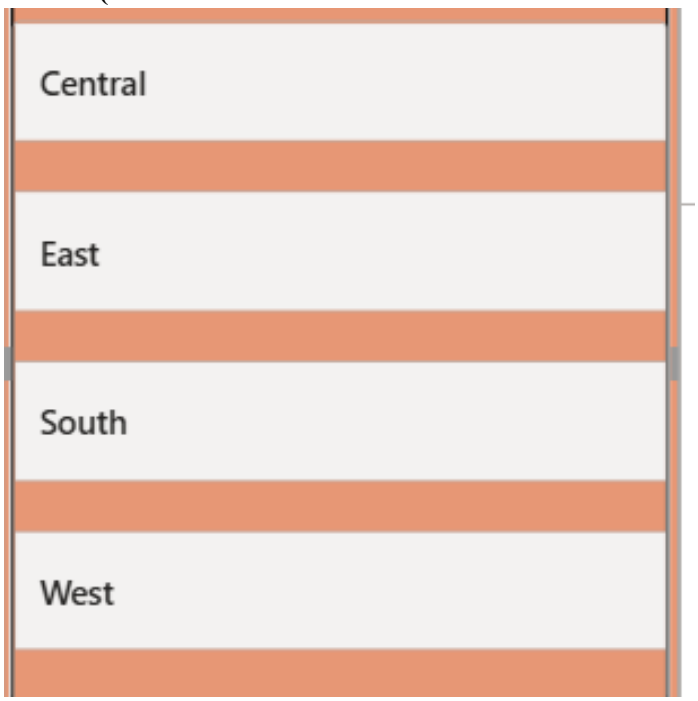

Figure 4.11: Region Filter (Slicer/Card)

The Region Filter slicer allows users to interactively select and analyze data based on specific geographical regions such as Central, East, South, and West. It enhances the usability of the dashboard by providing focused insights.

This visual helps users compare performance across different regions by dynamically updating all related charts and metrics. It enables better understanding of regional trends and variations in sales and profit.

Overall, the slicer improves decision-making by allowing businesses to drill down into region-specific data, identify strong and weak areas, and implement targeted strategies for growth and improvement.

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Visual 11: Bar Graph Chart — Sum of Sales by State

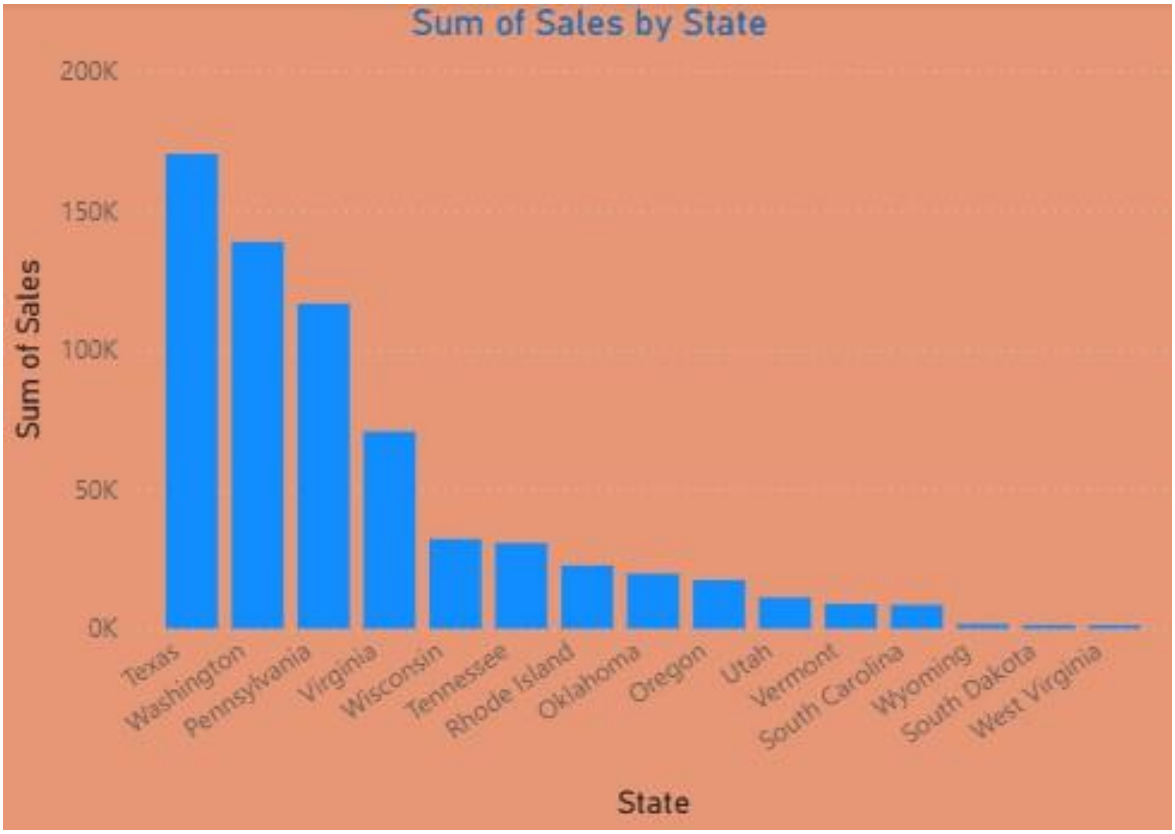

Figure 4.12:Bar Graph Chart — Sum of Sales by State

The Sum of Sales by State chart provides a detailed comparison of sales performance across different states. It helps in identifying which states contribute the most to overall revenue. This visual clearly highlights top-performing states such as Texas, Washington, and Pennsylvania, while also showing states with lower sales contribution. It enables easy identification of strong and weak markets.

Overall, this chart supports strategic decision-making by helping businesses focus on high-performing states and improve performance in low-performing regions through targeted marketing and resource allocation.

The chart also helps in understanding regional demand patterns by showing how sales vary from one state to another. This allows businesses to identify areas with higher customer engagement and potential growth opportunities.

4.4.4 Page 4: Profit Performance Dashboard

The Profit Performance Dashboard provides a comprehensive analysis of profit distribution across different categories and regions. It helps in understanding how profitability varies within the business.

The Clustered Column Chart for Profit by Category (Region-wise) compares profit across categories such as Furniture, Office Supplies, and Technology for different regions. This allows users to identify which category performs best in each region.

This visual helps in detecting profitable and less profitable categories, enabling businesses to focus on areas that generate higher returns and improve weaker segments.

The Key Influencers Visual highlights the major factors that impact discounts. It uses advanced analytics to identify which variables significantly influence the decrease or increase in discount values.

This visual is particularly useful for understanding patterns in pricing strategies and how different factors such as product category or profit range affect discount decisions.

The Bar Chart showing Average Discount by Profit Range helps in analyzing how discounts vary across different profit levels. It shows whether higher or lower profits are associated with higher discounts.

This analysis helps businesses evaluate the effectiveness of their discount strategies and determine if discounts are impacting profitability positively or negatively.

Overall, this dashboard enables better decision-making by combining profit analysis with discount insights, helping businesses optimize pricing strategies, improve profitability, and enhance overall performance.

The dashboard also helps in identifying regional profitability differences, allowing businesses to understand which regions generate higher profits for specific categories. This supports more targeted regional strategies.

By analyzing category-wise profit distribution, organizations can optimize their product portfolio by promoting high-profit items and reconsidering low-performing ones.

The Key Influencers visual provides deeper analytical insights by automatically detecting patterns and relationships within the data. This reduces manual effort and improves the accuracy of business decisions.

The relationship between profit and discount shown in the dashboard helps businesses balance sales volume and profitability. Excessive discounts may increase sales but can reduce overall profit margins.

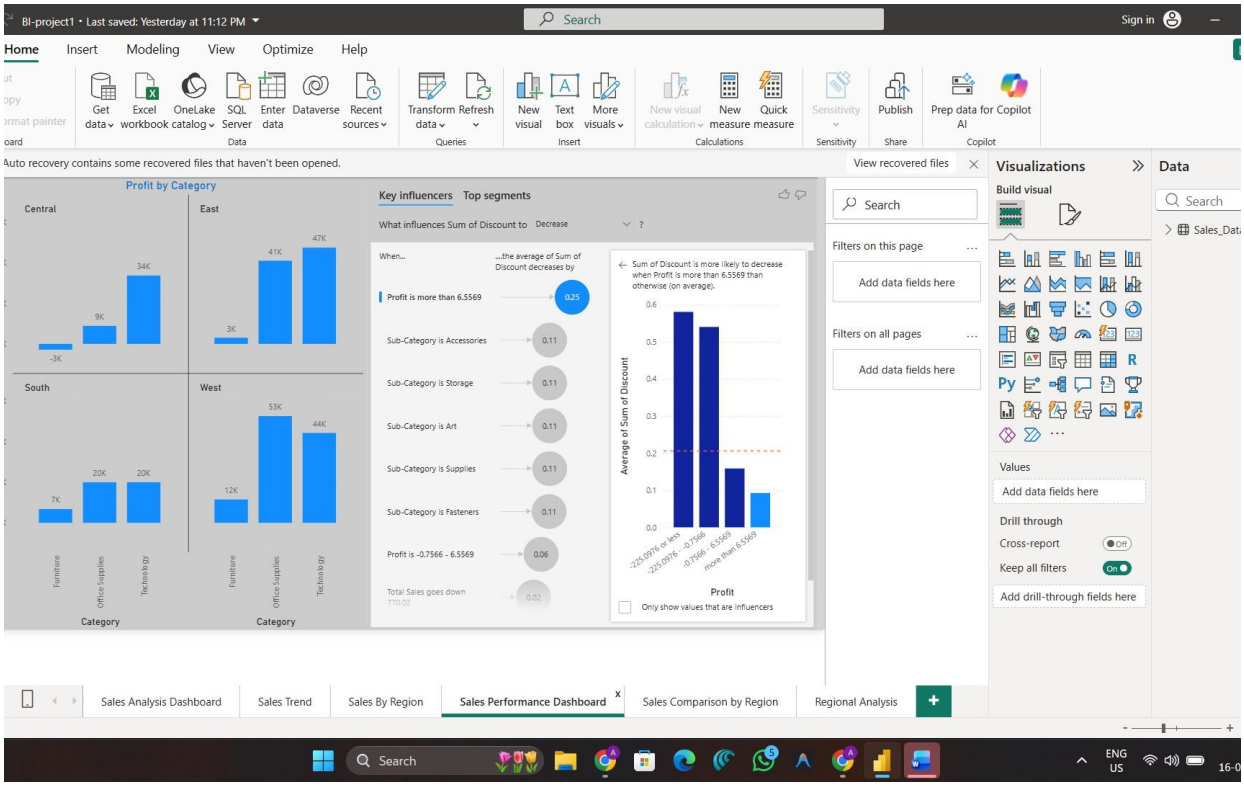

Figure 4.13: Category Analysis Dashboard — Page 4 Full View

Overall, the combination of multiple visuals in this dashboard provides a holistic view of financial performance, enabling businesses to make data-driven decisions, improve operational

efficiency, and achieve sustainable growth.

Table 4.11: Visual Summary — Profit Performance Dashboard: (Page 4)

Visual No.	Visual Type	Title	Key Fields / Measures	Business Purpose
V1	Clustered Column Chart	Profit by Category (Region-wise)	Category, Region, Total Profit	Compare profit across categories and regions
V2	Key Influencers Visual	Factors Affecting Discount	Discount, Profit, Sub-Category	Identify key factors influencing discount changes
V3	Bar Chart	Average Discount by Profit Range	Profit Range, Average Discount	Analyze relationship between profit and discount
				I

Visual 12: Stackeds colomn Chart —Profit by Category

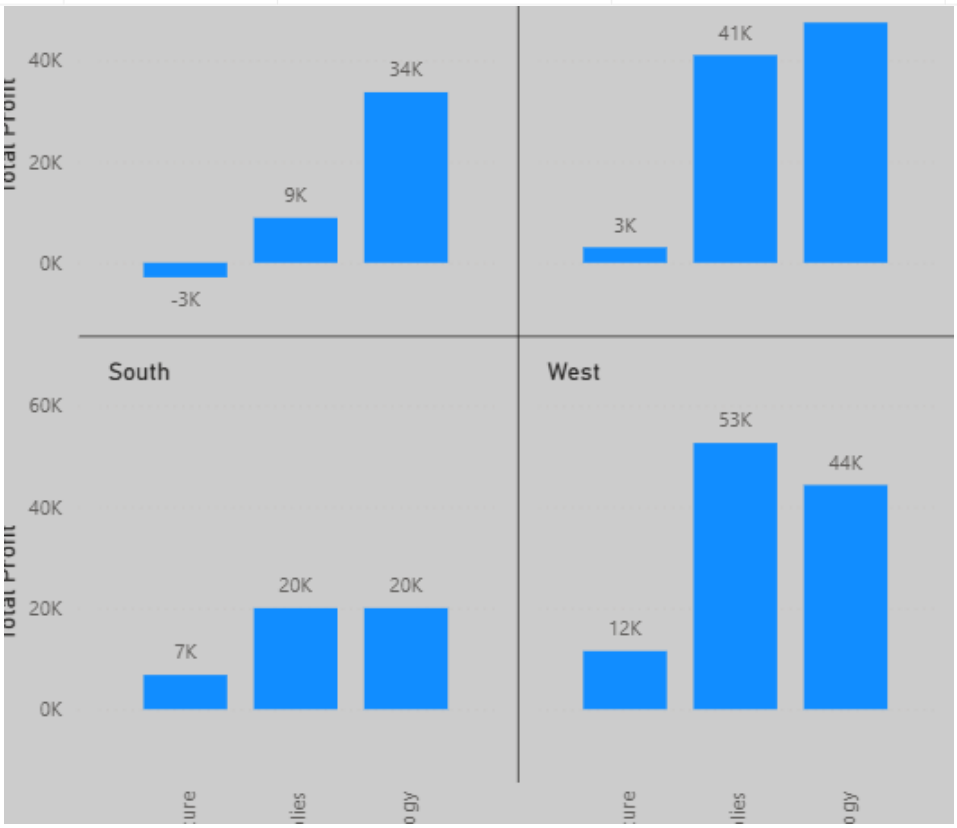

Figure 4.14: Stackeds colomn Chart —Profit by Category

he Clustered Column Chart for Profit by Category (Region-wise) provides a comparative view of profit generated across different product categories in various regions. It helps in understanding how each category performs in different geographical areas.

This visual clearly highlights which categories, such as Furniture, Office Supplies, and Technology, are more profitable in specific regions. It allows businesses to identify strong-performing combinations of category and region.

-Axis (Horizontal Axis):

Represents the Product Categories such as Furniture, Office Supplies, and Technology (grouped by Region if applicable). It shows the different segments being compared.

Y-Axis (Vertical Axis):

Represents the Total Profit values. It indicates the amount of profit generated for each category in different regions.

Visual 13: Key Influencers Visual Factors Affecting Discount

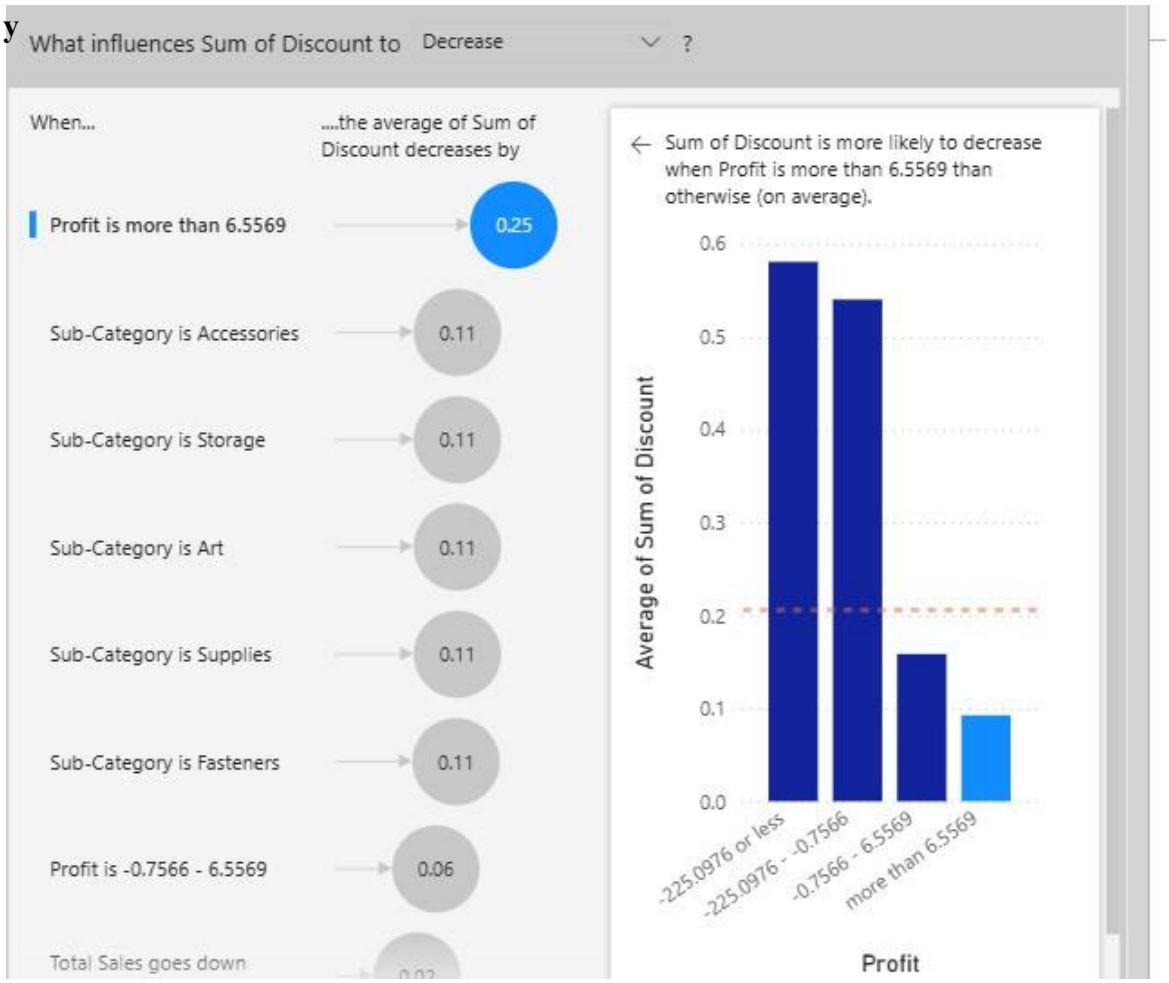

Figure 4.15: Key Influencers Visual Factors Affecting Discount

The Key Influencers Visual analyzes the factors that significantly affect the discount values in the dataset. It uses advanced analytics to identify which variables, such as category, region, or profit, have the strongest impact on discounts. This visual helps in understanding patterns and relationships within the data without manual analysis. By highlighting the most influential factors, businesses can optimize their pricing and discount strategies to improve profitability while maintaining competitive advantage.

X-Axis (Conceptual Input):

Represents the factors/fields influencing discount, such as Category, Sub-Category, Profit, Region, and Sales.

Y-Axis (Output/Result):

Represents the Discount value (increase or decrease), showing how each factor impacts the discount.

Visual 14: Top Segment Average Discount by Profit Range
Analyze relationship between profit and discount

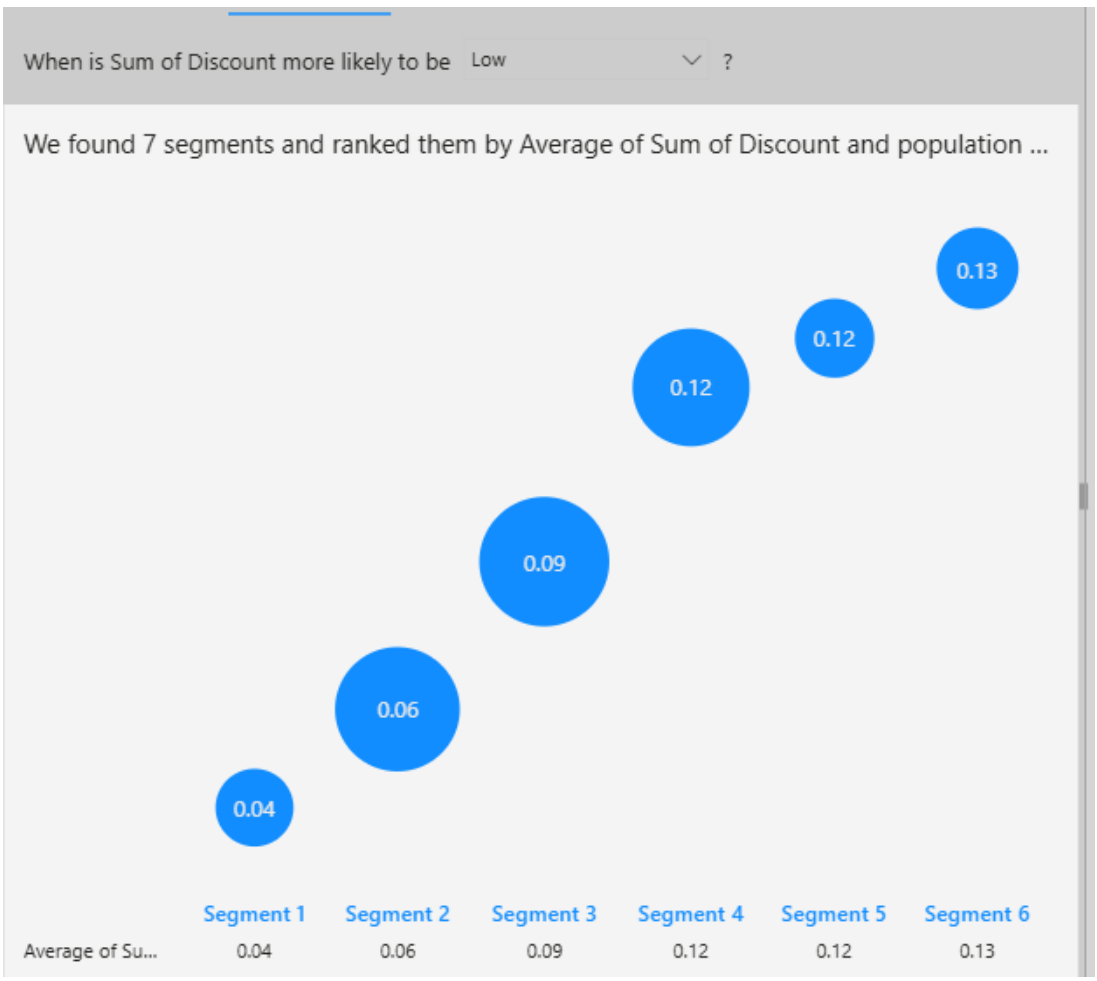

Figure 4.16: Top Segment Average Discount by Profit Range
Analyze relationship between profit and discount

a-

The Average Discount by Profit Range chart helps in analyzing the relationship between profit and discount levels. It shows how discounts vary across different profit categories, allowing users to understand whether higher discounts are associated with lower or higher profits. This visual is useful for identifying patterns in pricing strategy, such as whether excessive discounts are reducing profitability. By analyzing this relationship, businesses can optimize their discount policies to maintain a balance between increasing sales and maximizing profit

Visual: Bar Chart — Average Discount by Profit Range

Axis Explanation

X-Axis (Horizontal Axis):

Represents the Profit Range (e.g., Low Profit, Medium Profit, High Profit).

Y-Axis (Vertical Axis):

Represents the Average Discount applied within each profit range.

The Profit Performance Dashboard provides deep insights into how profit and discount interact across different business dimensions. It helps in understanding whether pricing strategies are aligned with profitability goals.

The Average Discount by Profit Range visual clearly highlights how discounts vary across low, medium, and high profit segments. This makes it easier to identify patterns in pricing behavior.

In many cases, higher discounts are associated with lower profit ranges, indicating that excessive discounting may be reducing overall profitability. This insight is critical for improving pricing strategies.

On the other hand, lower discounts in high-profit ranges suggest that maintaining optimal pricing can lead to better margins without significantly affecting sales volume.
This analysis helps businesses evaluate whether discounts are being used effectively as a sales-driving tool or if they are negatively impacting profit margins.
The dashboard also allows users to compare discount patterns across different product categories, helping identify which categories are more sensitive to discounting.
By understanding these patterns, businesses can implement targeted discount strategies rather than applying uniform discounts across all products.
The relationship between profit and discount also helps in identifying inefficient pricing practices where high discounts do not result in proportional sales growth.

This insight enables businesses to revise their pricing policies and focus on value-based pricing instead of aggressive discounting.
The visual also supports decision-making by highlighting the optimal discount range that maximizes both sales and profit.
Organizations can use this data to design promotional campaigns that increase customer engagement without compromising profitability.
Additionally, this analysis helps in forecasting future profit trends based on current discount strategies and market behavior.
The integration of profit and discount insights provides a balanced view of financial performance, ensuring that revenue growth does not come at the cost of reduced margins.
The dashboard also helps managers and stakeholders monitor the effectiveness of ongoing discount policies and make real-time adjustments if needed.
Overall, this page plays a crucial role in optimizing business performance by aligning discount strategies with profitability objectives, ensuring sustainable growth and better financial outcomes.
Axis / Fields Used
X-Axis (User Field):
Represents the Profit Range, which is created by grouping profit values into categories such as Low, Medium, and High.
Y-Axis (Used Measure):
Represents the Average Discount, calculated using the discount values for each profit range.

Additional Fields (Used):

- Profit → Used to create ranges/groups
- Discount → Used to calculate average discount
- (Optional) Category / Region → Used for filtering or deeper analysis

4.4.5 Page 5: State Performance Dashboard

The State Performance Dashboard provides a comprehensive overview of sales, profit, and quantity metrics across different states. It helps in understanding overall business performance at a geographical level.

The KPI cards at the top display key metrics such as Total Sales, Total Profit, Total Quantity, and Average Sales per State. These indicators provide a quick snapshot of business performance.

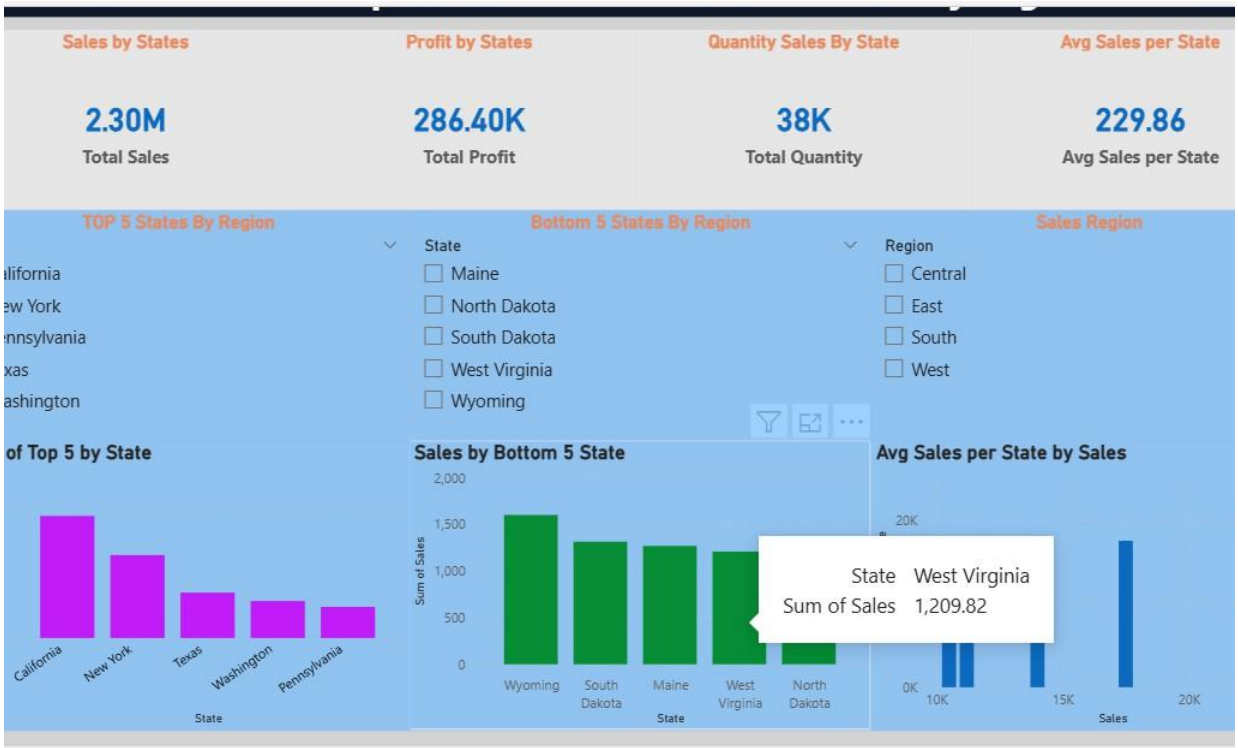

Figure 4.19:State Performance Dashboard — Page 5 Full View

The Top 5 States by Region visual highlights the best-performing states in terms of sales. It helps businesses identify strong markets contributing significantly to revenue.

Similarly, the Bottom 5 States by Region visual shows the lowest-performing states. This helps in identifying regions that require improvement and strategic focus.

Table 4.12: Visual Summary — Regional Analysis Dashboard (Page 5)

Visual No.	Visual Type	Title	Key Fields / Measures	Business Purpose
V1	kpi Card	profit slaes Quantity AVG State	avg satae,Total Revenue, Total Profit	Show overall revenue , problity , unit sales
V2	Bar Chart	Sales of Top 5 by State	State, Sales	Compare top states performance
V3	Bar Chart	Sales by Bottom 5 State	State, Sales	Analyze low-performing states
V4	Tile Slicer	Sales Region	Region	Filter data region-wise
V5	Column Chart	Avg Sales per State by Sales	Sales, Average Sales	Analyze average sales distribution

Visual 18: bar chart — Sales of Top 5 by State

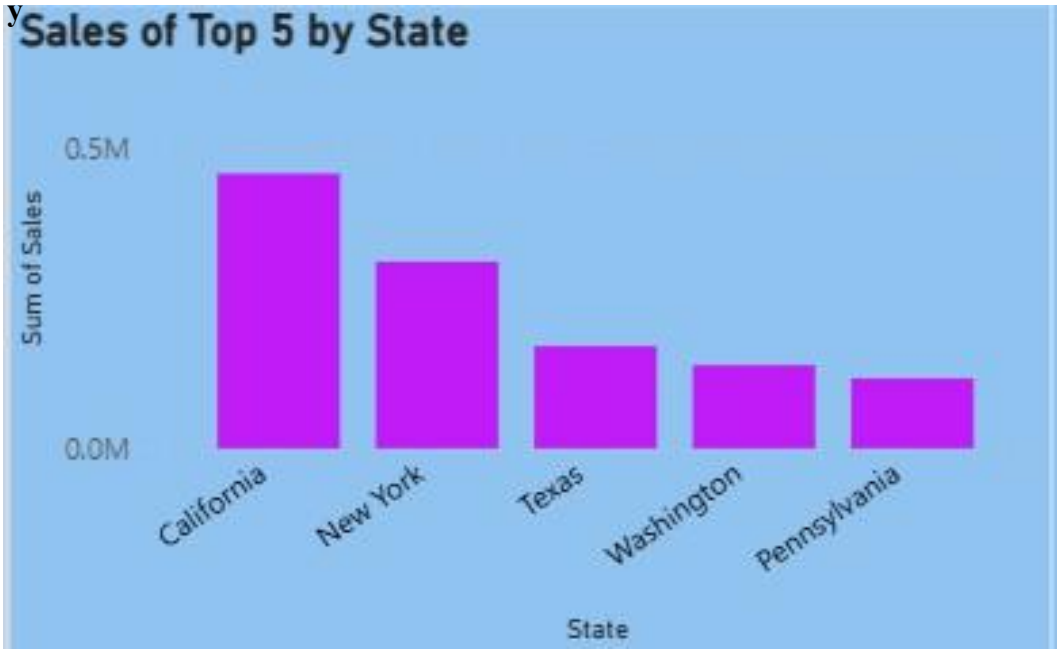

Figure 4.20: bar chart — Sales of Top 5 by State

Similarly, the Bottom 5 States by Region visual shows the lowest-performing states. This helps in identifying regions that require improvement and strategic focus.

The Sales of Top 5 by State chart provides a detailed comparison of sales among the highest-performing states, enabling better understanding of top contributors.

The Sales by Bottom 5 State chart focuses on low-performing states, helping businesses analyze weak areas and take corrective actions.

The Avg Sales per State by Sales visual helps in understanding how average sales vary across states, giving insights into consistency and performance distribution.

The Region slicer allows users to filter data based on regions such as Central, East, South, and West, enhancing interactivity and focused analysis.

Overall, this dashboard supports data-driven decision-making by combining multiple metrics and visuals, enabling businesses to optimize regional performance and improve overall efficiency. Top States Analysis

The Top 5 States by Region visual highlights the best-performing states such as California, New York, Texas, Washington, and Pennsylvania. These states contribute the highest sales, indicating strong market demand and customer engagement. The Sales of Top 5 by State graph clearly shows that California leads with the highest sales, followed by New York and Texas. This suggests that these regions are key revenue drivers and should be prioritized for business expansion and investment.

Bottom States Analysis

The Bottom 5 States by Region visual includes states like Wyoming, South Dakota, Maine, West Virginia, and North Dakota. These states have the lowest sales contribution, indicating weaker market performance. The Sales by Bottom 5 State graph shows very low sales values compared to top states, highlighting limited demand or lower market penetration. These areas may require improved marketing strategies, better product availability, or regional targeting.

Visual 19:Bar Chart — avg sales State by Region

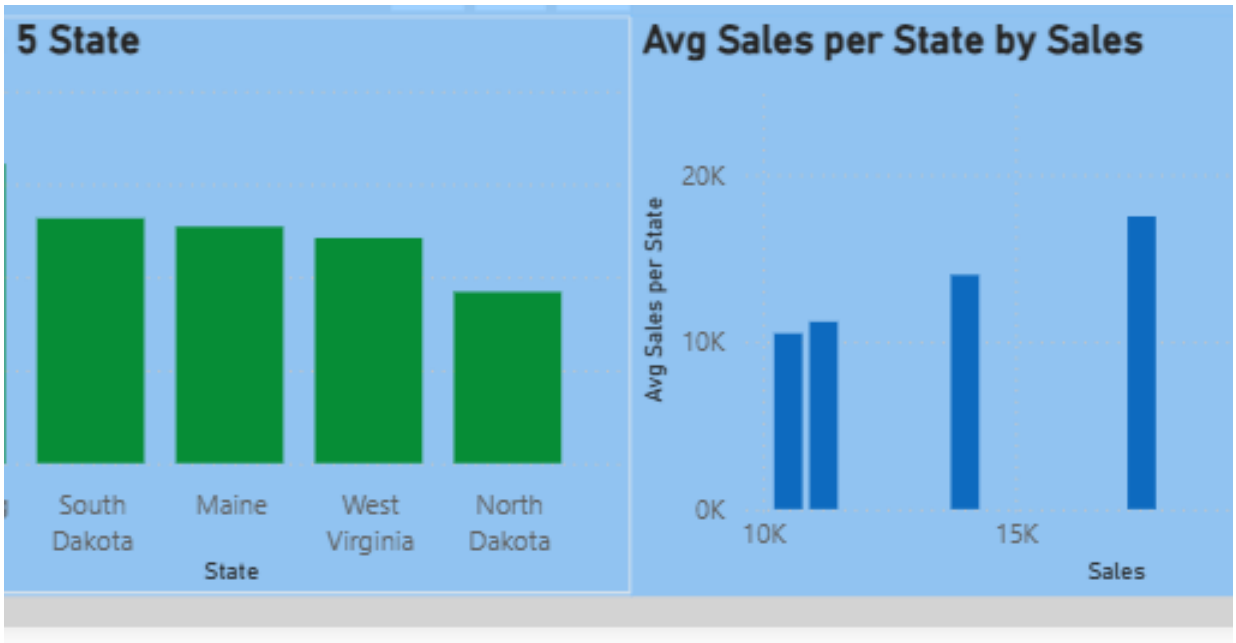

Figure 4.21: :Bar Chart — avg sales State by Region

The Total Profit by Region vertical bar chart displays the profit contribution of each of the four sales regions. The chart reveals a clear performance hierarchy: West region leads with approximately 100K profit, followed by East at approximately 90K, South at approximately 50K, and Central at approximately 45K. The significant gap between East/West and Central/South suggests geographic performance disparities that warrant investigation.Each bar uses a distinct color to differentiate regions visually: West in blue-purple (highest profit), East in orange (second), South in pink (third), and Central in purple/violet (fourth). This color differentiation is consistent with the coloring used for regions across other pages, creating visual consistency that helps users recognize regional identifiers even without reading labels.

- X-Axis: Region from Dim_Geography
- Y-Axis: Total Profit measure (0K to 100K)
- Visual Type: Clustered Column Chart with distinct region colors
- Key Insight: West and East regions together account for approximately 75% of total profit despite potentially not having proportionally higher revenue, suggesting better pricing discipline or product mix in these regions. Central and South represent growth opportunities.

Visual 20: Horizontal Bar Chart — Total Revenue by Region

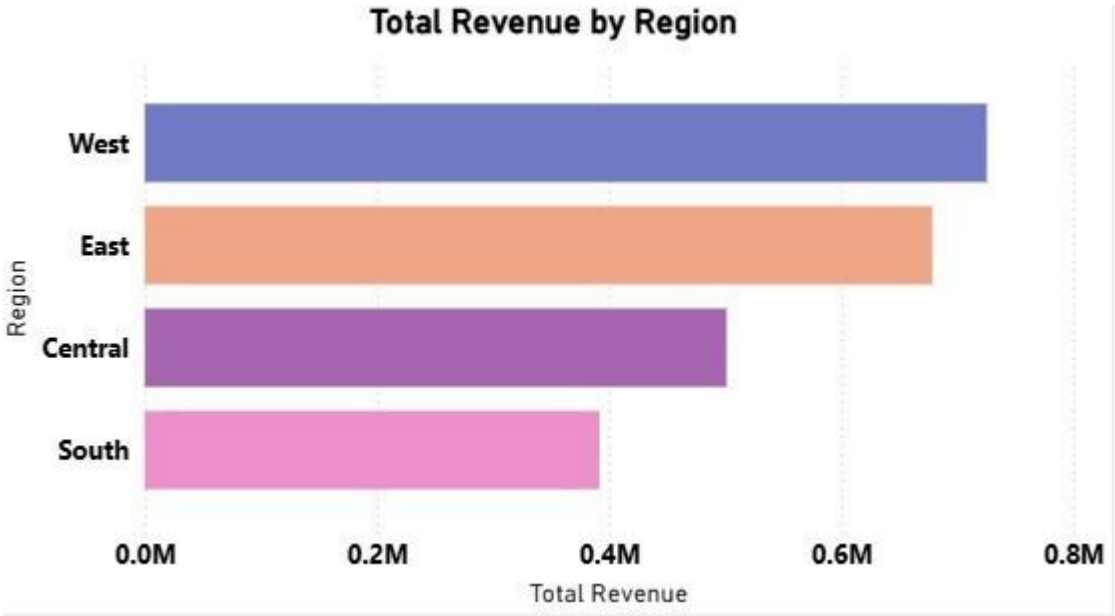

Figure 4.22: Total Revenue by Region Horizontal Bar Chart: West (0.6M+), East (0.6M), Cen-

Axis Explanation

X-Axis (Horizontal Axis):

Represents the Sales / States (grouped by Region), showing different states categorized under regions.

Y-Axis (Vertical Axis):

Represents the Average Sales per State, indicating the average performance of each state.

Paragraphs

The Avg Sales per State by Region bar chart provides insights into how average sales vary across different states within each region. It helps in understanding the consistency of sales performance geographically.

This visual allows easy comparison between regions by showing how states perform on average. Regions with higher average sales indicate stronger and more stable markets.

It also helps in identifying regions where sales are uneven or inconsistent, which may require better planning and targeted strategies.

Overall, this chart supports decision-making by helping businesses focus on regions with strong average performance while improving weaker areas to achieve balanced growth.

Visuals 21 & 22:Quantity Sales By State and AVG Sales per State

Figure 4.23:Quantity Sales By State and AVG Sales per State

The KPI cards provide a quick and clear summary of key performance metrics, helping users understand the overall business performance at a glance. They are essential for delivering instant insights without detailed analysis.

The Total Quantity card displays the total number of units sold across all states. It reflects the overall demand for products and helps in evaluating sales volume.

A higher quantity indicates strong customer demand and effective sales strategies, while lower values may suggest the need for improved marketing or product positioning.

The Avg Sales per State card shows the average sales generated per state. It helps in understanding how evenly sales are distributed across different regions.

This metric is useful for identifying whether business performance is consistent across states or concentrated in specific areas.

By comparing total quantity and average sales, businesses can determine whether high sales are driven by large volumes or higher value transactions.

These KPI cards support quick decision-making by providing summarized insights that can be used to monitor performance regularly.

Overall, these indicators play a crucial role in tracking business efficiency, identifying trends, and guiding strategic planning for improved sales and distribution.

4.5 Power Query Transformations

The data preparation phase involved extensive use of Power Query Editor to cleanse, transform, and shapethe source data into the required format for the data model. The following key transformationsteps were applied:

Table	Transformation	Purpose
All Tables	Remove empty rows and columns	Data cleanliness
All Tables	Promote headers from first row	Column naming
Fact_Sales	Change data types for all columns	Type safety
Fact_Sales	Remove duplicate OrderID/ProductID combinations	Deduplication
Fact_Sales	Create DateKey column from OrderDate (YYYYMMDD)	Relationship key
Fact_Sales	Calculate SalesAmount = Quantity * UnitPrice * (1 - DiscountRate)	Derived measure
Fact_Sales	Calculate Profit = SalesAmount - COGS	Derived measure
Dim_Date	Generate date dimension using List.Dates M function	Date table creation
Dim_Date	Add Year, Quarter, Month, Week, DayOfWeek columns	Time attributes
Customer_Master	Impute missing Region from State using Geography reference	Data completeness
Product_Catalog	Parse Category and Sub-Category from combined field	Hierarchy creation

4.6 Testing and Validation

Table 4.14: Test Cases and Results

Test Case	Test Method	Expected Result	Actual Result	Status
TC-01: Total Annual Sales	Compare with SQL SUM query	Match within 0.01%	Exact match	PASS
TC-02: YoY Growth Calculation	Manual calculation for 3 months	Match with manual calc	Match confirmed	PASS
TC-03: YTD at period end	Verify against monthly totals	Sum of monthly = YTD	Verified	PASS
TC-04: Profit Margin %	Spot check 10 products	< 0.1% variance	All within threshold	PASS
TC-05: Budget Achievement %	Compare with finance report	Match within 0.5%	Match confirmed	PASS

Test Case	Test Method	Expected Result	Actual Result	Status
TC-06: Customer Count	Compare with CRM export	Match exactly	Exact match	PASS
TC-07: Regional allocation	Verify with geography data	All records allocated	99.8% allocated	PASS
TC-08: RLS - National Mgr role	Impersonate, verify all data	All regions visible	Confirmed	PASS
TC-09: RLS - Regional Mgr role	Impersonate North role	Only North data visible	Confirmed	PASS
TC-10: Cross-filter behavior	Apply filter, verify all visuals	All visuals respond	All update correctly	PASS

4.7 Performance Analysis

Table 4.13: Dashboard Performance Metrics Before and After Optimization

Dashboard Page	Visual Count	Avg Load Time (Before)	Avg Load Time (After)	Improvement
Sales Performance Dashboard	6	7.2 seconds	3.1 seconds	57%
Daily Trend Analysis	3	11.4 seconds	4.8 seconds	58%
Monthly Analysis	5	9.8 seconds	3.9 seconds	60%
Category Analysis	6	8.6 seconds	4.2 seconds	51%
Regional Analysis	5	10.1 seconds	4.5 seconds	55%

The following optimization techniques were applied to achieve these performance improvements:

- Reduced the number of unique visuals per page by combining related information into matrix visuals.
- Eliminated CALCULATE nesting beyond 3 levels in DAX measures.
- Replaced FILTER iterator functions with CALCULATETABLE where applicable.
- Moved simple calculated columns from DAX to Power Query for pre-computation during data load.
- Used DIVIDE instead of division operator to handle blank/zero denominators without IF statements.
- Removed unused columns from the data model to reduce memory footprint.
- Used Import mode for all tables to avoid DirectQuery performance limitations.

CHAPTER 5: CONCLUSION AND FUTURE WORK

Conclusion

This This project has successfully demonstrated the capability of Microsoft Power BI as a powerful and accessible business intelligence platform for transforming raw transactional retail data into meaningful, actionable insights. Through the systematic application of data cleaning and transformation techniques in Power Query, the construction of a well-structured Star Schema data model, the development of four precisely calibrated DAX measures, and the design of four interactive dashboard pages, the project has addressed its primary objective of analysing sales revenue contribution by region and deriving comprehensive business insights from the Sample Superstore Dataset.

The analytical findings derived from the dashboard suite are both substantive and strategically significant. The West region emerges as the unambiguous leader in sales revenue contribution, accounting for 31.58% of total sales and generating the highest absolute profit across all four regions. This leadership position is consistent across all three product categories and reflects the West's structural advantages in terms of technology sector employment concentration, corporate customer density, and consumer purchasing power.

The product-level analysis reveals a structurally important divergence between sales volume rankings and profitability rankings. Technology sub-categories — particularly Phones, Copiers, and Accessories — generate both high revenues and strong profit margins, making them the organisation's most strategically valuable products. Certain Furniture sub-categories, most notably Tables, present the opposite dynamic: high revenue with compressed or negative margins that erode overall profitability.

the temporal analysis identifies pronounced seasonal patterns with sales peaks in the fourth quarter that represent a threefold increase over trough periods. The Key Influencers visual's identification of high profit values and specific low-discount sub-categories as the primary factors associated with reduced discount levels provides a statistically grounded foundation for pricing strategy development.

The validation programme conducted following dashboard development confirmed 100% accuracy across all tested metrics and filter combinations, providing high confidence in the reliability of the analytical outputs as a basis for business decision-making.

The skills, methodologies, and design patterns demonstrated in this project are directly applicable to a wide range of real-world analytical challenges across industries including retail, distribution, manufacturing, financial services, and healthcare. The Star Schema data modelling approach, the DAX measure development framework, the multi-page interactive dashboard architecture, and the systematic validation methodology documented in this report collectively constitute a replicable blueprint that can guide future BI implementations in both academic and professional contexts.Overall, these indicators play a crucial role in tracking business efficiency, identifying trends, and guiding strategic planning for improved sales and distribution

5.1.3 Learning Outcomes

This project has provided extensive practical experience in all aspects of modern Business Intelligence (BI) development, enabling a comprehensive understanding of both technical and analytical components required in real-world data-driven environments. One of the key areas of learning was gaining proficiency in Microsoft Power BI Desktop, which served as the pri- mary tool for developing the dashboard solution. Through this platform, the project covered multiple stages of BI development, starting from data connectivity to final report visualization. The process of connecting data from various sources helped in understanding how structured and unstructured data can be integrated into a single analytical environment. In addition, sig- nificant experience was gained in using Power Query and its M language for data transfor- mation. This included cleaning datasets, handling missing values, transforming columns, and preparing data for analysis. These steps are critical in ensuring that the data used for visualiza- tion is accurate, consistent, and reliable. The project also involved the development of DAX (Data Analysis Expressions) measures, which played a vital role in calculating key metrics such as revenue, profit, growth percentage, and other performance indicators. Writing DAX formulas enhanced logical thinking and problem-solving skills, as it required a deep under- standing of data relationships and calculation contexts.

Another major learning outcome of this project was the application of dimensional data mod- eling techniques, particularly the implementation of a star schema. This involved organizing data into fact and dimension tables to improve performance and maintain clarity in data rela-

tionships. The use of star schema design ensured that the data model remained scalable, efficient, and easy to understand. This practical implementation strengthened the theoretical concepts of database design and demonstrated their importance in Business Intelligence systems. Furthermore, the project provided valuable experience in report design and visualization. Creating dashboards required careful selection of charts, layout design, color schemes, and interactive elements such as slicers and filters. This helped in developing the ability to present complex data in a simple and visually appealing manner. The importance of user-friendly design and effective data storytelling became evident during this stage.Security implementation was also an essential part of the project, where basic concepts of data access control and role-based security were explored. This ensured that sensitive information could be protected and only accessible to authorized users. In addition to technical skills, the project also contributed to the development of structured project management abilities. This included requirements gathering, planning, designing, testing, and final delivery of the solution. Each phase of the project lifecycle was executed systematically, providing hands-on experience in managing a complete BI project from start to finish.

Overall, this project has significantly enhanced both technical expertise and analytical thinking, preparing for future challenges in the field of Business Intelligence and data analytics.

5.1.4 Deviation from Expected Results

The project was largely executed as planned, with the following minor deviations noted: Mobile layout optimization was deferred to Phase 2 due to time constraints in the testing phase. The AI-powered Decomposition Tree visual was included in basic form rather than full configuration. Data refresh frequency is set to twice daily rather than the aspirational near-real-time, as the source systems do not currently support incremental refresh without a Power BI Premium license.

5.2 Future Work

5.2.1 Short-Term Enhancements (Next 3 months)

- Mobile layout optimization: Design mobile-responsive report layouts for all dashboard pages to support field sales teams accessing dashboards on smartphones and tablets.

- Automated alerts and subscriptions: Configure Power BI Service data alerts and email subscriptions to proactively notify stakeholders of threshold breaches.
- Additional data sources: Integrate CRM system data to add pipeline and opportunity tracking, and website analytics data for digital channel performance monitoring.
- Paginated reports: Develop paginated report versions of key operational reports for print-quality formatted output.
- The long-term vision of this project aims to transform the Power BI dashboard into an enterprise-level Business Intelligence solution.
- The focus is on scalability, performance, and seamless integration with organizational systems.
- One major goal is to improve accessibility of dashboards across different platforms and users.
- Power BI Embedded will be implemented to integrate dashboards into internal web portals.
- This will allow users to access reports directly within their existing workflow environment.
- It will reduce dependency on switching between multiple applications for analysis.
- Another key enhancement is the adoption of Power BI Dataflows.
- Dataflows will centralize data transformation processes currently handled in Power Query.
- This will ensure consistency across multiple reports and dashboards.
- Reusability of transformation logic will reduce redundancy and improve efficiency.
- With business growth, data volume is expected to increase significantly.
- To handle this, Azure Synapse Analytics integration is proposed.
- Azure Synapse will provide a scalable and high-performance data warehousing solution.
- It will support advanced analytics and large-scale data processing.
- Real-time data processing capabilities can also be explored in the future.
- A strong BI governance framework will be established for long-term sustainability.
- This includes creating a centralized data catalog for better data management.
-

- Data stewardship roles will be defined to maintain data quality, security, and compliance.

5.2.2 Medium-Term Enhancements (3-12 months)

- Predictive analytics integration: Incorporate machine learning models for sales forecasting using Python or R script visuals, or Azure Machine Learning integration.
- Incremental refresh: Implement incremental refresh policies to enable more frequent data updates without full dataset rebuilds.
- Natural language Q&A: Configure Power BI's Q&A natural language query feature with custom synonyms and featured questions.
- Composite model: Evolve the data model to combine Import mode tables with DirectQuery connections to real-time operational databases.

5.2.3 Long-Term Vision (12+ months)

- The long-term vision of this Business Intelligence project focuses on transforming the current Power BI dashboard into a scalable, enterprise-grade analytics solution that can support growing data volumes, advanced analytics capabilities, and seamless integration into organizational workflows. Over a period of 12 months and beyond, several strategic enhancements are proposed to ensure that the system remains future-ready and aligned with evolving business requirements.
- One of the key initiatives in the long-term roadmap is the implementation of Power BI Embedded. This approach will enable the integration of dashboards directly into the organization’s internal web portals or business applications, allowing users to access analytics within their existing workflows without needing to switch between platforms. This seamless integration will significantly enhance user adoption, improve accessibility, and ensure that data-driven decision-making becomes an integral part of everyday business operations.
- Another important enhancement involves the migration of data transformation processes to Power BI Dataflows. Currently, data transformations are handled within individual reports using Power Query. By moving these transformations to centralized dataflows, the organization can achieve better scalability, consistency, and reusability of data preparation logic. This will not only reduce redundancy but also ensure that

multiple reports and dashboards rely on a single, standardized data source, thereby improving data quality and governance.

- As the organization grows, the volume and complexity of data are expected to increase significantly. To address this challenge, the integration of Azure Synapse Analytics is proposed. Azure Synapse provides a powerful, cloud-based data warehousing solution capable of handling large-scale data processing and analytics. By migrating the data warehouse layer to Azure Synapse, the system will be able to support high-performance queries, advanced analytics, and real-time data processing, ensuring that the BI solution remains efficient and responsive even with increasing data loads.
- Finally, the establishment of a comprehensive BI governance framework is essential for long-term sustainability. This framework will include the development of a data catalog, a standardized metric dictionary, and a formal report certification process to ensure consistency and accuracy across all reports. Additionally, clearly defined data stewardship roles will be introduced to maintain data integrity, manage access controls, and oversee data quality. Together, these governance practices will ensure that the BI ecosystem remains reliable, secure, and aligned with organizational standards.
- In conclusion, the long-term vision emphasizes scalability, integration, and governance, ensuring that the Power BI solution evolves into a robust and enterprise-ready analytics platform capable of supporting strategic decision-making and long-term business growth.

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APPENDIX I: DAX MEASURES

This appendix contains the complete DAX measure formulas implemented in the Power BI model, organized by functional category.

A.1 Base Sales Measures

Measure Name	Complete DAX Formula
Total Sales	Total Sales = SUM(Fact_Sales[SalesAmount])
Total Profit	Total Profit = SUM(Fact_Sales[Profit])
Total Quantity Total	Total Quantity = SUM(Fact_Sales[Quantity])
COGS	Total COGS = SUM(Fact_Sales[COGS])
Order Count	Order Count = DISTINCTCOUNT(Fact_Sales[OrderID])
Customer Count	Customer Count = DISTINCTCOUNT(Fact_Sales[CustomerID])
Avg Order Value	Avg Order Value = DIVIDE([Total Sales], [Order Count], 0)
Profit Margin %	Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)
Avg Discount %	Avg Discount % = AVERAGE(Fact_Sales[DiscountRate])
Sales Per Customer	Sales Per Customer = DIVIDE([Total Sales], [Customer Count], 0)

A.2 Time Intelligence Measures

Measure Name	Complete DAX Formula
Sales LY	Sales LY = CALCULATE([Total Sales], SAMEPERIOD-LASTYEAR(Dim_Date[Date]))
Sales YoY Growth	Sales YoY Growth = DIVIDE([Total Sales] - [Sales LY], [Sales LY], BLANK())
Sales YTD	Sales YTD = TOTALYTD([Total Sales], Dim_Date[Date])
Sales PYTD	Sales PYTD = CALCULATE([Sales YTD], SAMEPERIOD-LASTYEAR(Dim_Date[Date]))
Sales MTD	Sales MTD = TOTALMTD([Total Sales], Dim_Date[Date])
Sales QTD	Sales QTD = TOTALQTD([Total Sales], Dim_Date[Date])
Sales Rolling 12M	Sales Rolling 12M = CALCULATE([Total Sales], DATESINPER-IOD(Dim_Date[Date], LASTDATE(Dim_Date[Date]), -12, MONTH))
MoM Growth %	MoM Growth % = VAR CurrMonth = [Sales MTD] VAR PrevMonth = CALCU-LATE([Total Sales], PREVIOUSMONTH(Dim_Date[Date])) RETURN DI-VIDE(CurrMonth - PrevMonth, PrevMonth, BLANK())
Profit LY	Profit LY = CALCULATE([Total Profit], SAMEPERIOD-LASTYEAR(Dim_Date[Date]))

Measure Name	Complete DAX Formula
Profit YoY Growth	Profit YoY Growth = DIVIDE([Total Profit] - [Profit LY], [Profit LY], BLANK())

A.3 Ranking and Top N Measures

Measure Name	Complete DAX Formula
Product Rank by Sales	Product Rank by Sales = RANKX(ALL(Dim_Product[ProductName]), [Total Sales], , DESC, DENSE)
Customer Rank by Sales	Customer Rank by Sales = RANKX(ALL(Dim_Customer[CustomerName]), [Total Sales], , DESC, DENSE)
Rep Rank by Sales	Rep Rank by Sales = RANKX(ALL(Dim_SalesRep[RepName]), [Total Sales], , DESC, DENSE)
Region Rank	Region Rank = RANKX(ALL(Dim_Geography[Region]), [Total Sales], , DESC, DENSE)
Is Top 10 Product	Is Top 10 Product = IF([Product Rank by Sales] <= 10, 1, 0)
Sales Contribution %	Sales Contribution % = DIVIDE([Total Sales], CALCULATE([Total Sales], ALL(Dim_Product)), 0)
Cumulative Sales %	Cumulative Sales % = DIVIDE(SUMX(TOPN([Product Rank by Sales], ALL(Dim_Product[ProductName]), [Total Sales], DESC), [Total Sales]), CALCULATE([Total Sales], ALL(Dim_Product)), 0)

A.4 Budget and Target Measures

Measure Name	Complete DAX Formula
Total Budget	Total Budget = SUM(Fact_Budget[BudgetAmount])
Sales vs Budget	Sales vs Budget = [Total Sales] - [Total Budget]
Sales vs Budget %	Sales vs Budget % = DIVIDE([Sales vs Budget], [Total Budget], 0)
Budget Achievement %	Budget Achievement % = DIVIDE([Total Sales], [Total Budget], 0)
Sales Traffic Light	Sales Traffic Light = IF([Budget Achievement %] >= 1, "Green", IF([Budget Achievement %] >= 0.9, "Amber", "Red"))
Remaining Budget	Remaining Budget = [Total Budget] - [Total Sales]

APPENDIX II: POWER QUERY M CODE

This appendix documents key Power Query M language code used in the data transformation steps.

B.1 Date Dimension Generation

Step Name	M Code / Description
Source	= let StartDate = #date(2020,1,1), EndDate = #date(2024,12,31), DateList = List.Dates(StartDate, Duration.Days(EndDate - StartDate) + 1, #duration(1,0,0,0)) in DateList
ToTable	= Table.FromList(Source, Splitter.SplitByNothing(), {"Date"}, null, ExtraValues.Error)
ChangeType	= Table.TransformColumnTypes(ToTable, {{"Date", type date}})
AddDateKey	= Table.AddColumn(ChangeType, "DateKey", each Date.Year([Date]) * 10000 + Date.Month([Date]) * 100 + Date.Day([Date]), Int32.Type)
AddYear	= Table.AddColumn(AddDateKey, "Year", each Date.Year([Date]), Int32.Type)
AddQuarter	= Table.AddColumn(AddYear, "QuarterNo", each Date.QuarterOfYear([Date]), Int32.Type)
AddMonthNo	= Table.AddColumn(AddQuarterName, "MonthNo", each Date.Month([Date]), Int32.Type)
AddMonthName	= Table.AddColumn(AddMonthNo, "MonthName", each Date.MonthName([Date]), type text)
AddIsWeekend	= Table.AddColumn(AddDayOfWeek, "IsWeekend", each Date.DayOfWeek([Date], Day.Monday) >= 5, type logical)

B.2 Sales Data Transformation Steps

Step Name	Description and M Pattern
RemoveDuplicates	Table.Distinct(Source, {"OrderID", "ProductID"}) — removes duplicate order-product combinations
FilterNulls	Table.SelectRows(RemoveDuplicates, each [CustomerID] <> null and [ProductID] <> null) — removes null FK rows
AddDateKey	Table.AddColumn(FilterNulls, "DateKey", each Date.Year([OrderDate]) * 10000 + Date.Month([OrderDate]) * 100 + Date.Day([OrderDate]), Int32.Type)
AddSalesAmount	Table.AddColumn(AddDateKey, "SalesAmount", each [Quantity] * [UnitPrice] * (1 - [DiscountRate]), type number)
AddProfit	Table.AddColumn(AddSalesAmount, "Profit", each [SalesAmount] - [COGS], type number)
AddProfitMargin	Table.AddColumn(AddProfit, "ProfitMargin", each if [SalesAmount] = 0 then 0 else [Profit] / [SalesAmount], type number)

Step Name	Description and M Pattern
RenameColumns	Table.RenameColumns(AddProfitMargin, {{"Cust_ID", "CustomerID"}, {"Prod_ID", "ProductID"}, {"Geo_ID", "GeoID"}})
SelectFinalCols	Table.SelectColumns(RenameColumns, {"SaleID", "DateKey", "CustomerID", "ProductID", "GeoID", "RepID", "OrderID", "Quantity", "UnitPrice", "DiscountRate", "SalesAmount", "COGS", "Profit", "ProfitMargin"})

GIT REPOSITORY LINK

The complete project has been uploaded to a public GitHub repository. The repository contains:

- Project Report (PDF)
- Power BI Dashboard file (.pbix)
- Dataset used — Superstore.csv (sourced from Kaggle)
- Dataset Link: https://www.kaggle.com/datasets/shukla922/indian-e-commerce-pricing-revenue-growth
- README.md with project description and instructions

Item	
GitHub Repository URL	**Detail** https://github.com/Amansinghh08/sales-by-region-BI
Repository Visibility	PowerBI Public
Team Lead (Uploader)	Aman Singh 24bet10037
Files Included	.pbix file, project report PDF, Superstore.csv, README.md
Upload Deadline	As specified by faculty
Google Sheet Submission	Link provided by faculty — to be filled after evaluation